

# Basic Manufacturing Processes

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ME 204

by

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National Institute of Technology Karnataka, Surathkal



# Course Content

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Introduction to Manufacturing

Metal Casting

Metal Forming

Metal Joining

# Course Content

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Introduction, Classification, Gas Welding, Arc Welding, Resistance welding, Submerged Arc welding, Equipment details and working of Gas Metal Arc Welding (TIG & MIG), Carbon Arc Welding, Advanced Welding processes, Welding defects and inspection. Friction and friction stir welding, EBW, LBW, Dissimilar metal joining, Welding codes, Welding qualification, Fatigue of welded joints.  
Hands-on Training on Different Welding Machines

# Evaluation Plan

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|                      |                                              |
|----------------------|----------------------------------------------|
| Internal Assessment: | (20%) – [Tests/Quizzes/Assignments/Seminars] |
| Mid Semester Exam:   | (30%)                                        |
| End Semester Exam:   | (50%)                                        |

## Reference Books:

- Parmar, R.S, Welding processes and Technology, Khanna Publishers, 1997.
- Richard L. Little, Welding & Welding Technology, McGraw Hill, 1973.

# Manufacturing



# What is Manufacturing

# Manufacturing

(Derived from Latin word)

Manus

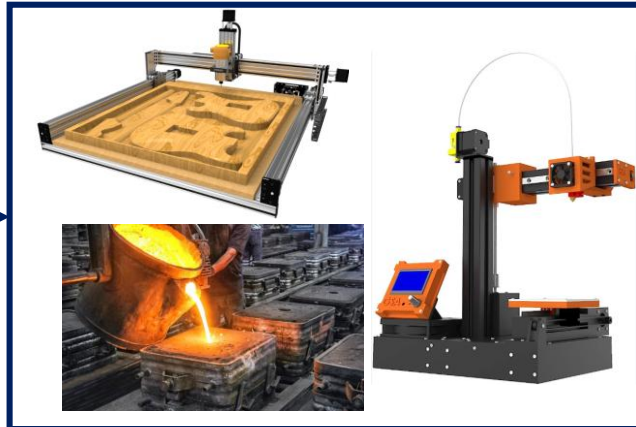
Hand

Factus

Made



Raw  
Materials

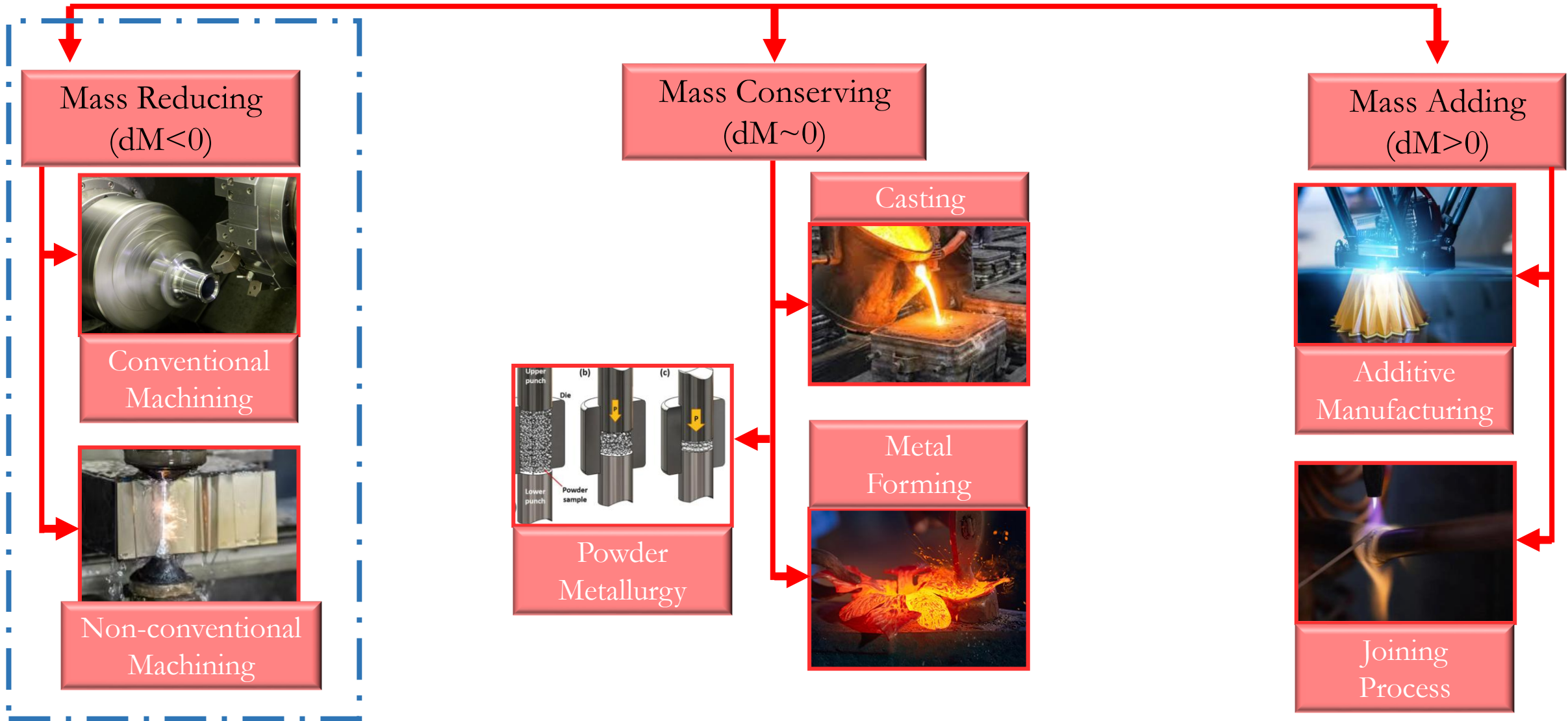


Manufacturing  
process



Useful  
Products

## Manufacturing

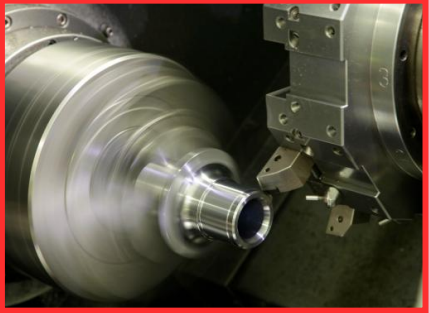


# Manufacturing: Mass Reducing

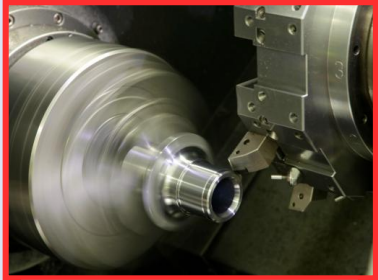
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Mass Reducing  
( $dM < 0$ )



Conventional  
Machining



Turning



Milling



Drilling



Grinding



## 5

[illegible]

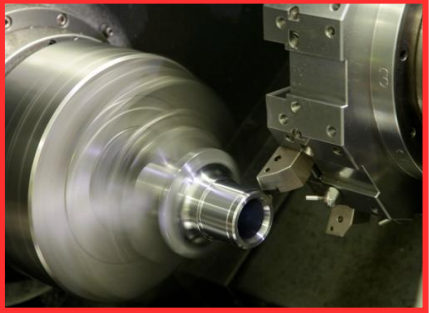
# Complex Geometries and $\mu$ -Machining

# Manufacturing: Mass Reducing

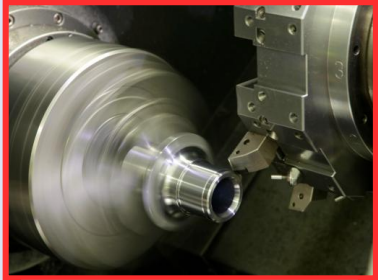
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Mass Reducing  
( $dM < 0$ )



Conventional  
Machining



Turning



Milling



Drilling



Grinding



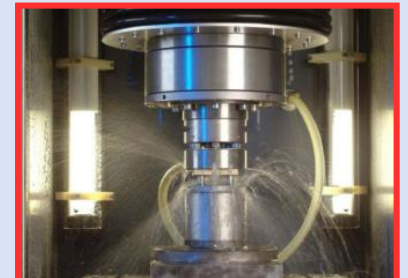
Non-conventional  
Machining



Mechanical



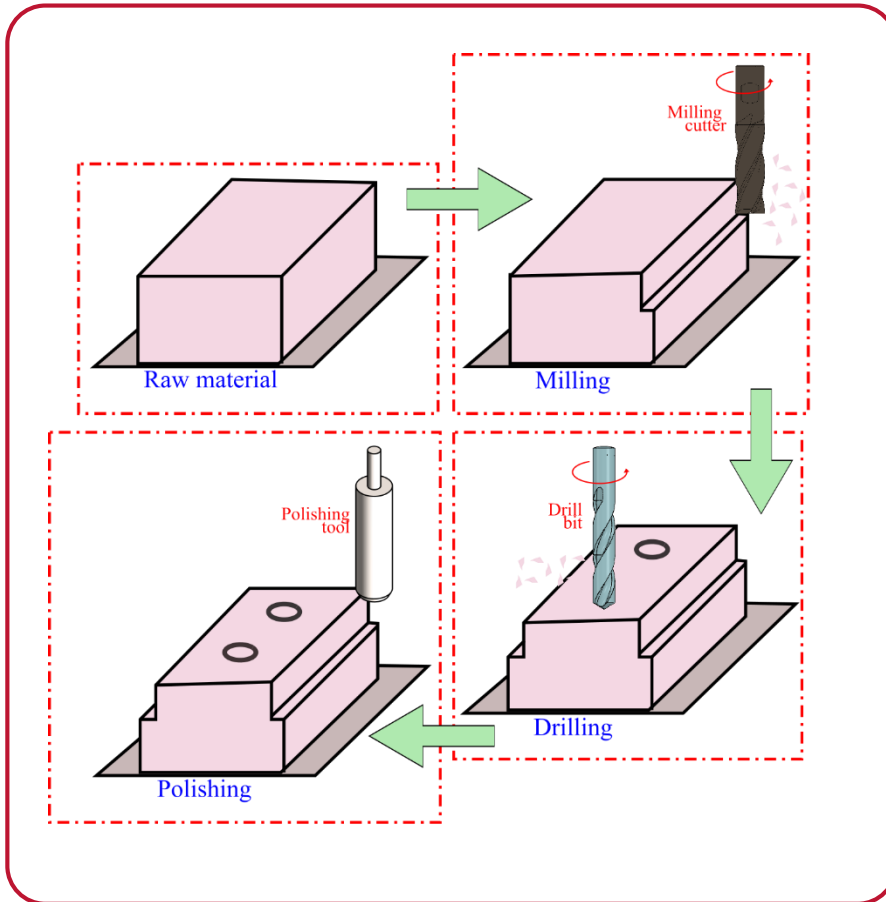
Thermal



Chemical & Electrochemical

# Introduction

## Subtractive Manufacturing





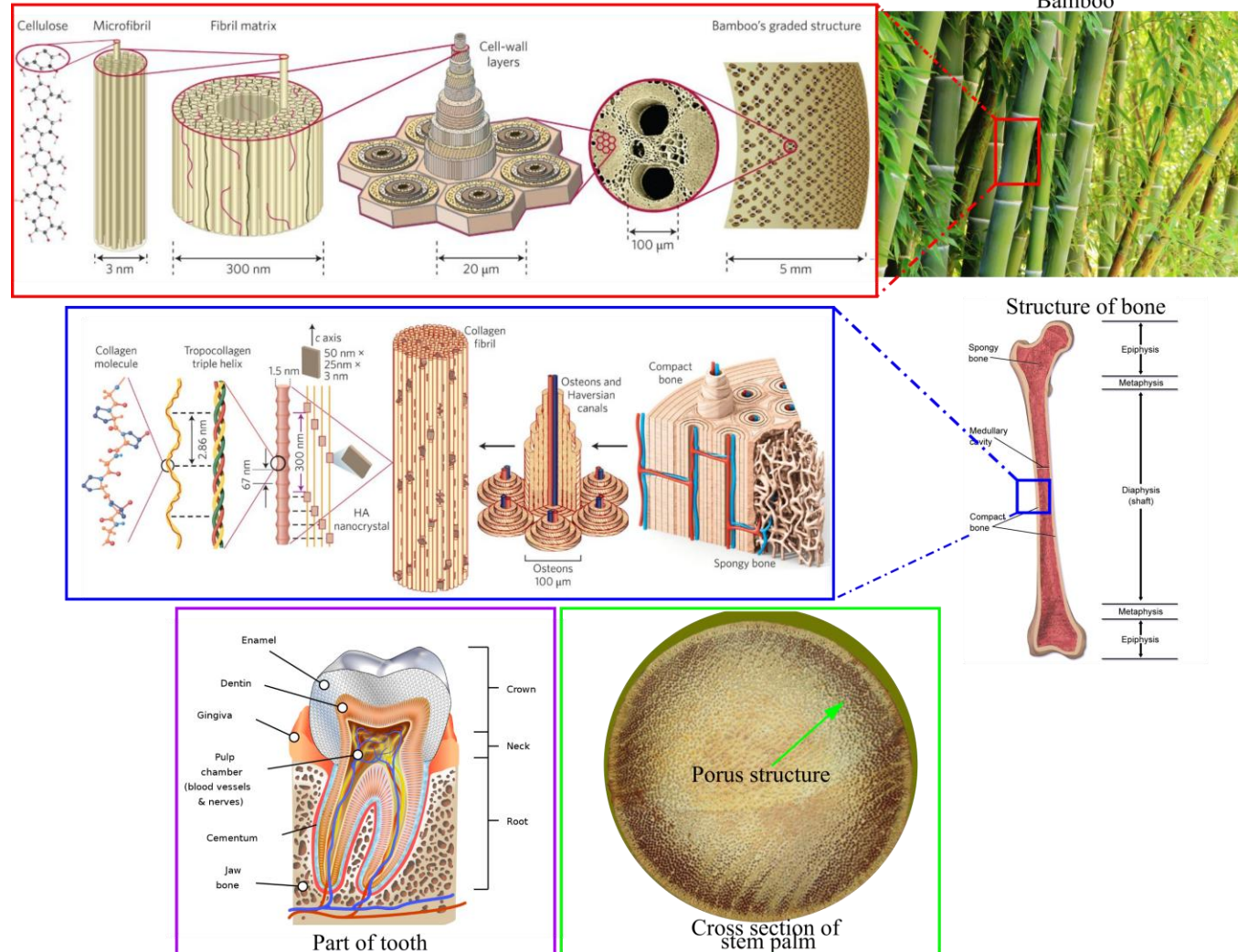
# Introduction

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# Introduction

## How Nature manufactures Close to Nature ! Flat ? Symmetric ? Homogeneity ?



# Introduction

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## How Nature manufactures

Nature does not make individual parts and assembly





# Introduction

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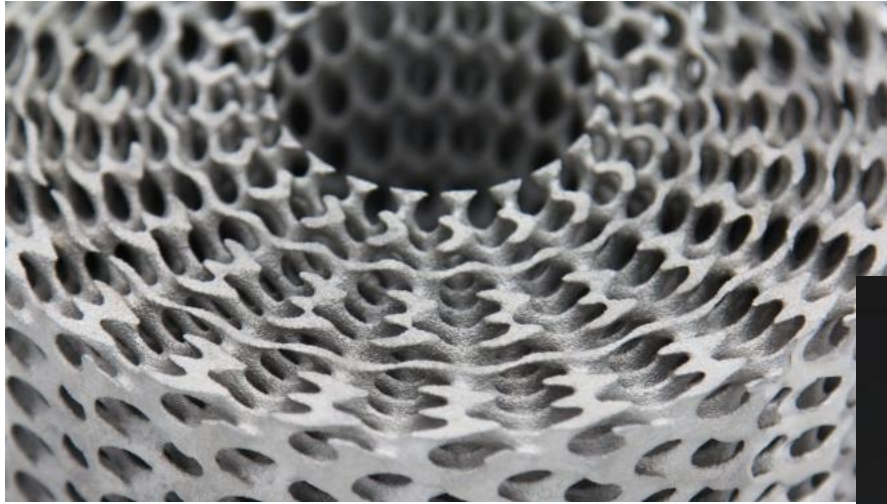
## How Nature manufactures

Nature's manufacturing has no wastage!



# Introduction

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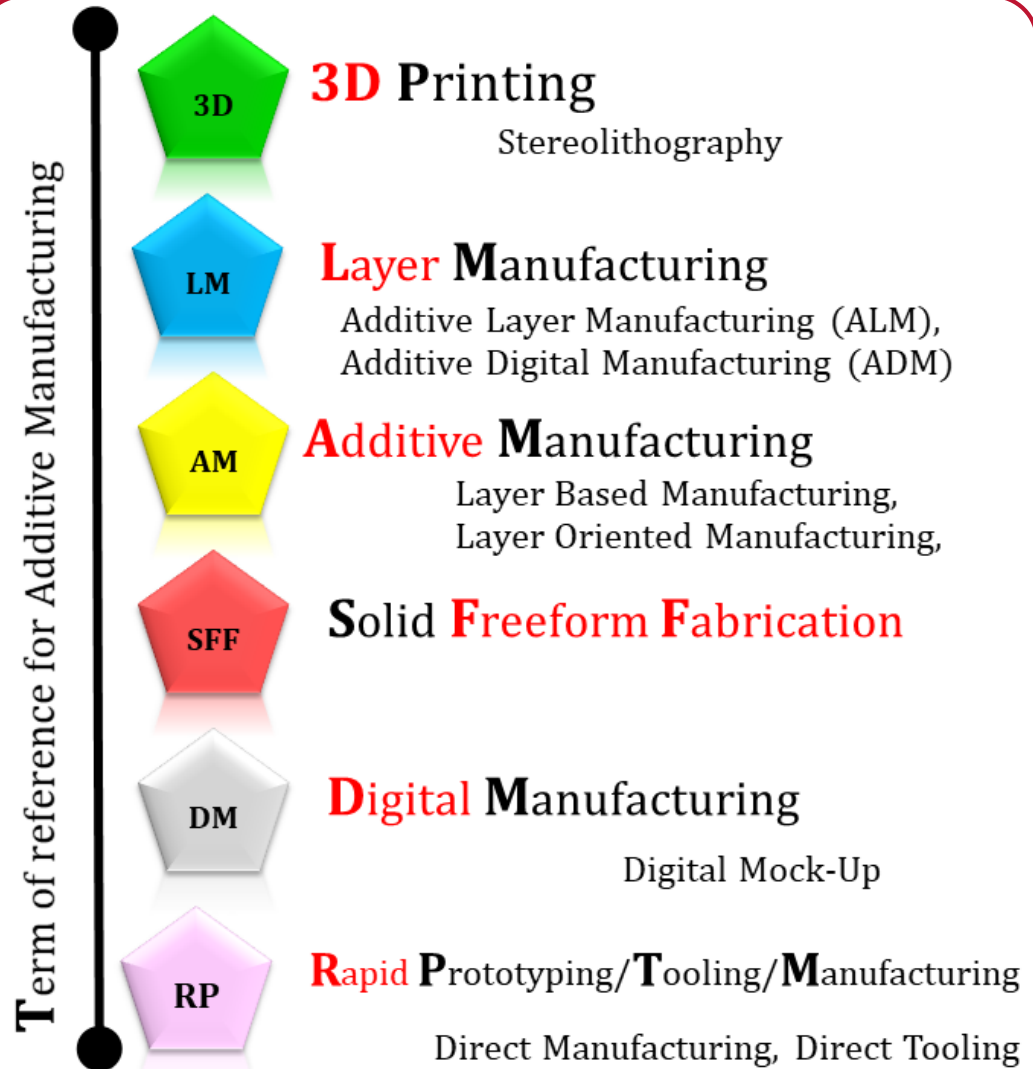
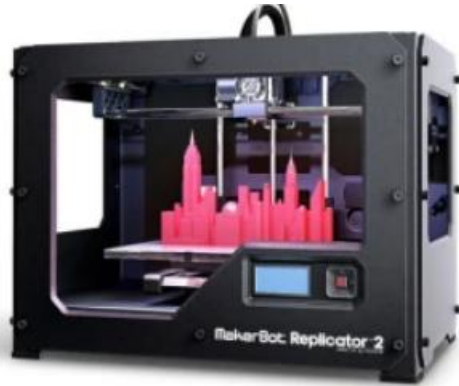
# Introduction

## 3D Printing/Additive Manufacturing



# Introduction

## 3D Printing/Additive Manufacturing



# Introduction

## 3D Printing/Additive Manufacturing



# Introduction

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**Pre-process(tessellate, slice, support)**



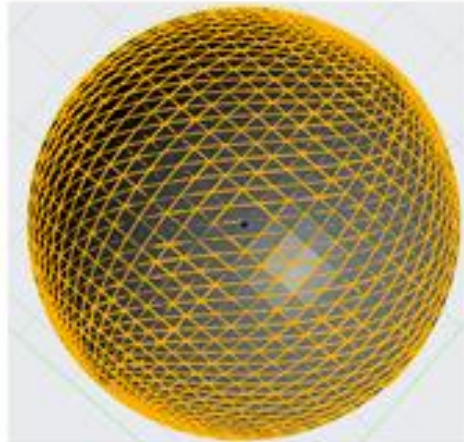
**Build part  
(additive/stacking)**



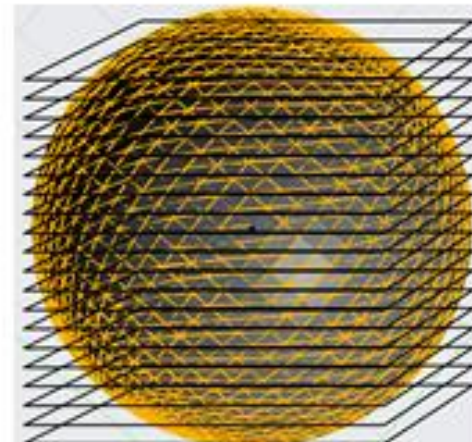
**Post-process  
(break support, finish)**



CAD file



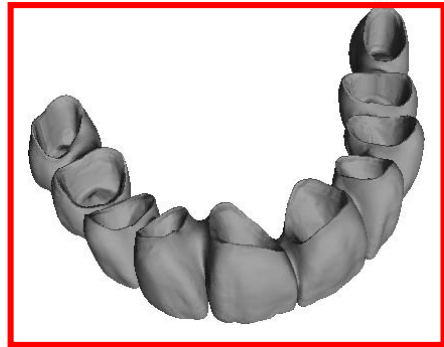
STL file



Sliced file

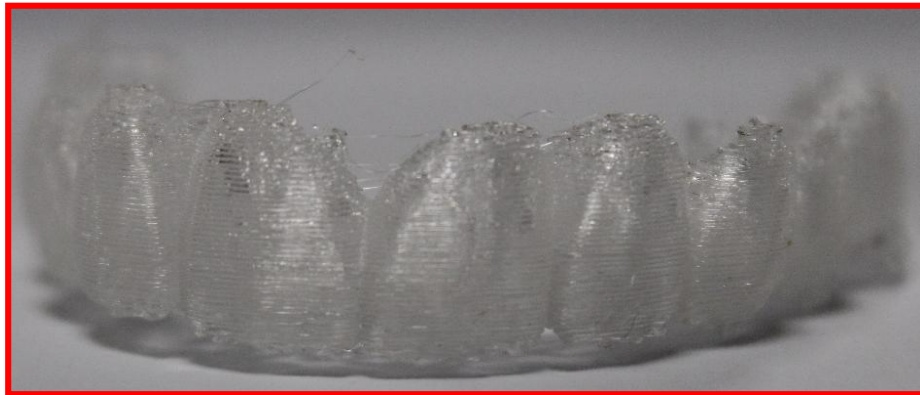
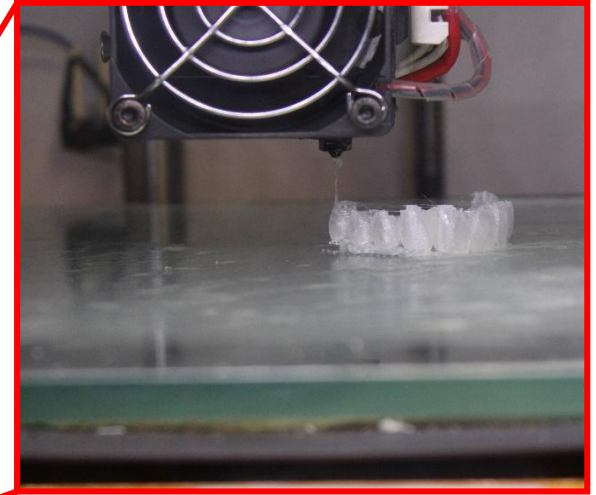
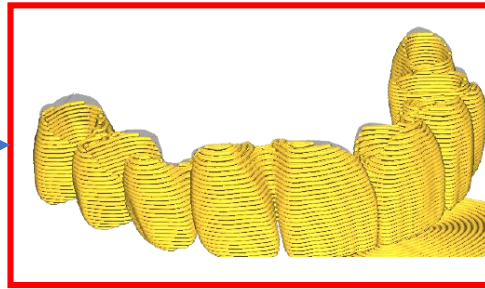


# Introduction

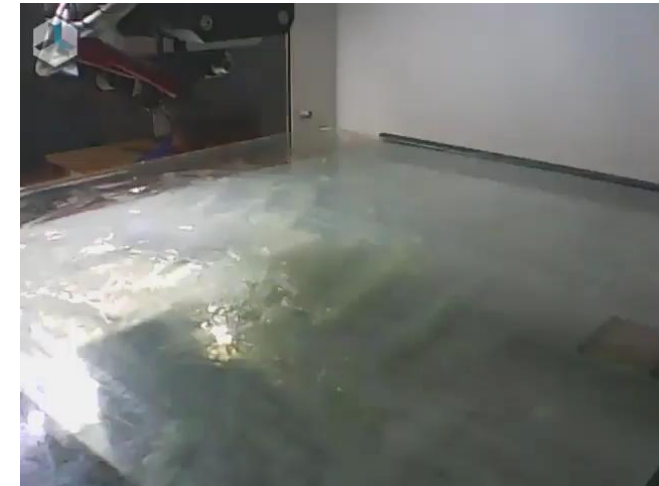


CAD model

Slicing



Final Product



# Metal Joining Process

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- There are a lot of ways to join two materials together.
- In some cases, adhesives like glues or cements might work.
- Fasteners like staples or rivets or screws or nails can create a pretty strong bond. However, when you are trying to join metal parts together, none of the above is likely to work.
- To adequately join metal parts together you need to solder, braze or weld the two metals
  - all processes based on heat

# Overview – Joining Processes

**P** = Permanent

**NP** = Non-permanent

| Metallurgical | Mechanical            | Adhesive        |
|---------------|-----------------------|-----------------|
| Brazing P     | Screw Fasteners NP    | Natural P       |
| Soldering P   | Compression Joints NP | Elastomer P     |
| Welding P     | Riveting P            | Thermoplastic P |
|               | Sheet Metal Folding P | Thermoset P     |

# Welding

- **Welding** is a permanent joining process in which two pieces of metal join together to form one piece by heating the metals to their melting points
- Additional metal also called filler metal is added during the heating process to help bond the two pieces together
- In general, it is a process in which two metal pieces similar (or) dissimilar may be joined by heating them to a temperature high enough to fuse the metals with (or) without the application of pressure and with (or) without the aid of filler material

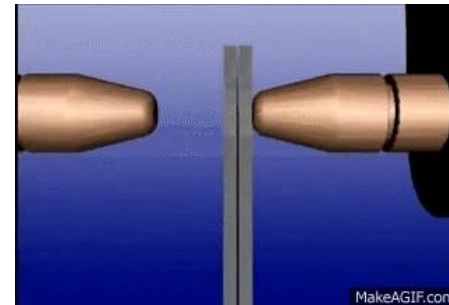
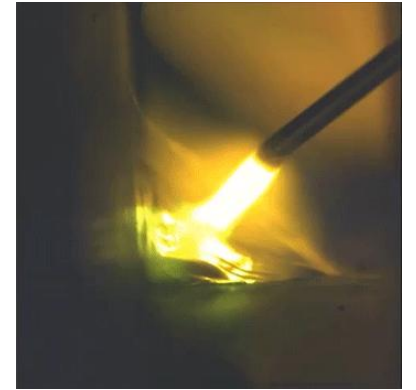
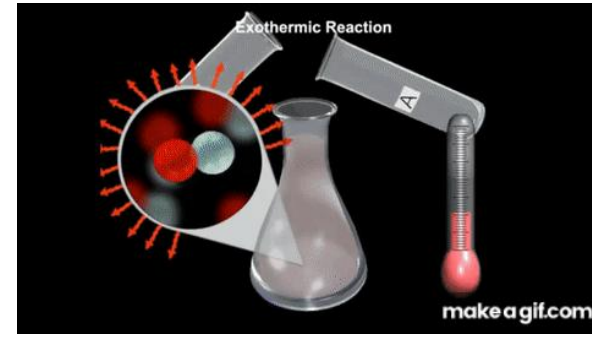
- The filler metal is supplied to form the joint, either from the electrode itself (or) by filler material.
- The temperature of the heat produced is of the order of 6000° to 7000°C.



# Welding

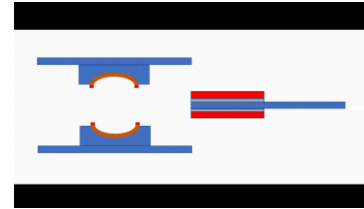
Heat may be obtained by

1. Chemical reaction
2. Electric arc
3. Electrical resistance
4. Frictional heat



# Types of welding

**Plastic welding:** metals to be joined are to be heated to the plastic state and then forced together by external pressure without the addition of filler material. Eg. forge welding, resistance welding...etc

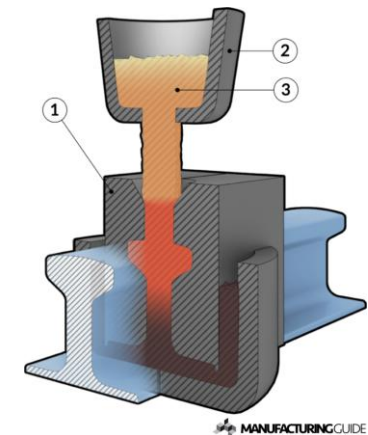
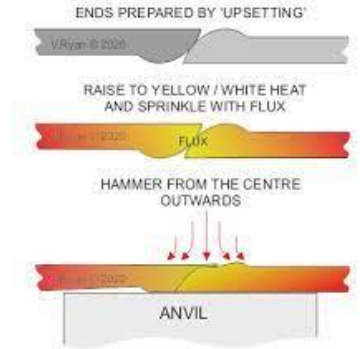
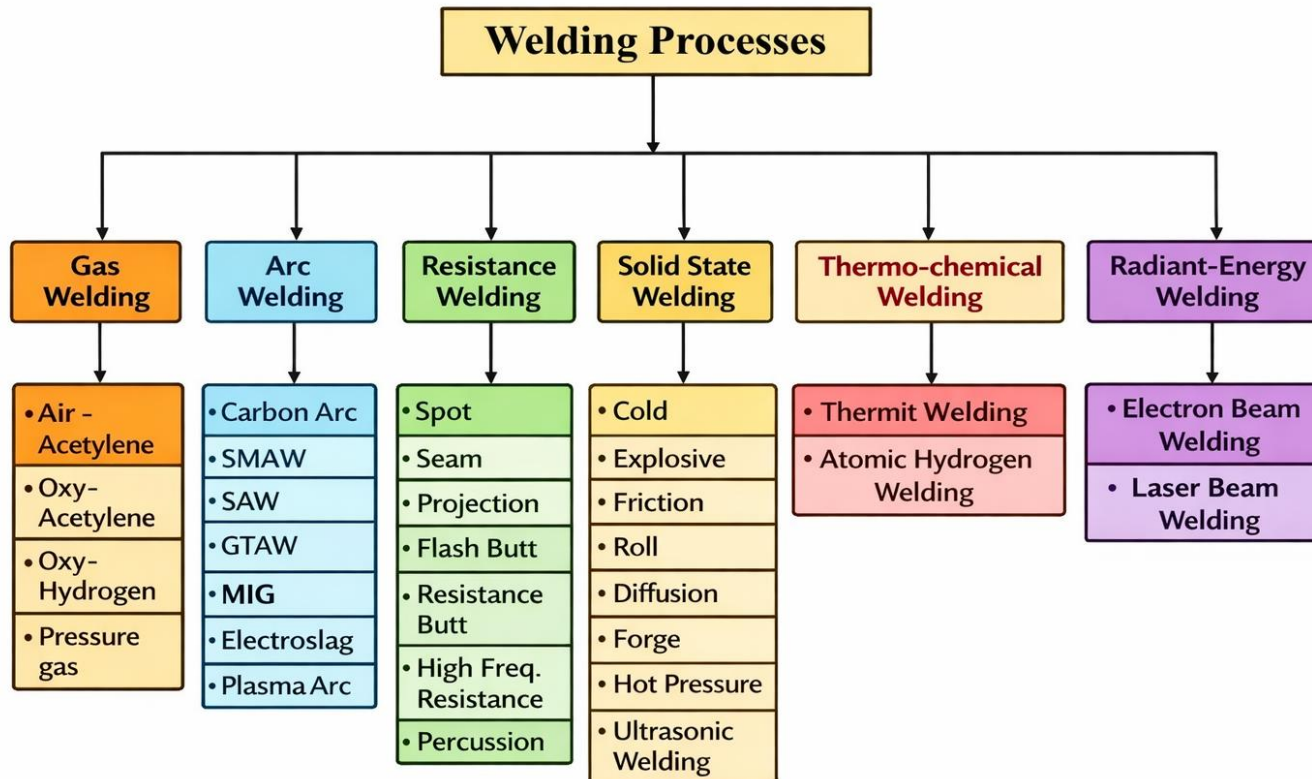


**Fusion welding:** no pressure is involved but a very high temperature is produced in or near the joint. The metal at the joint is heated to the molten state and allowed to solidify. A filler material may be used during the welding process. Eg. Oxy-acetylene welding, carbon arc welding...etc



# Types of welding

## Classification of Welding Processes



**Autogenous:** Solid State    **Homogenous:** Liquid State    **Heterogeneous:** Mixed State

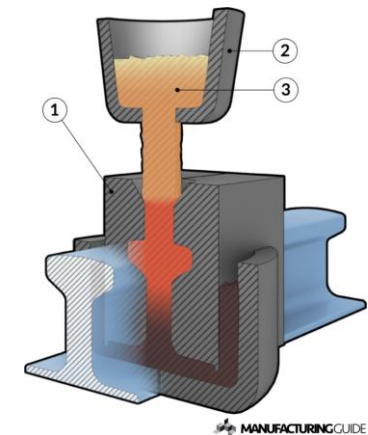
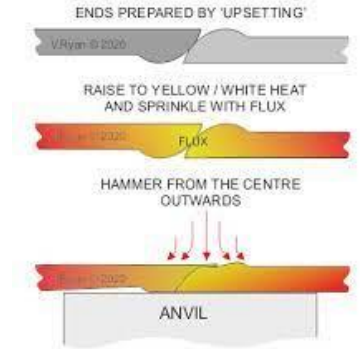
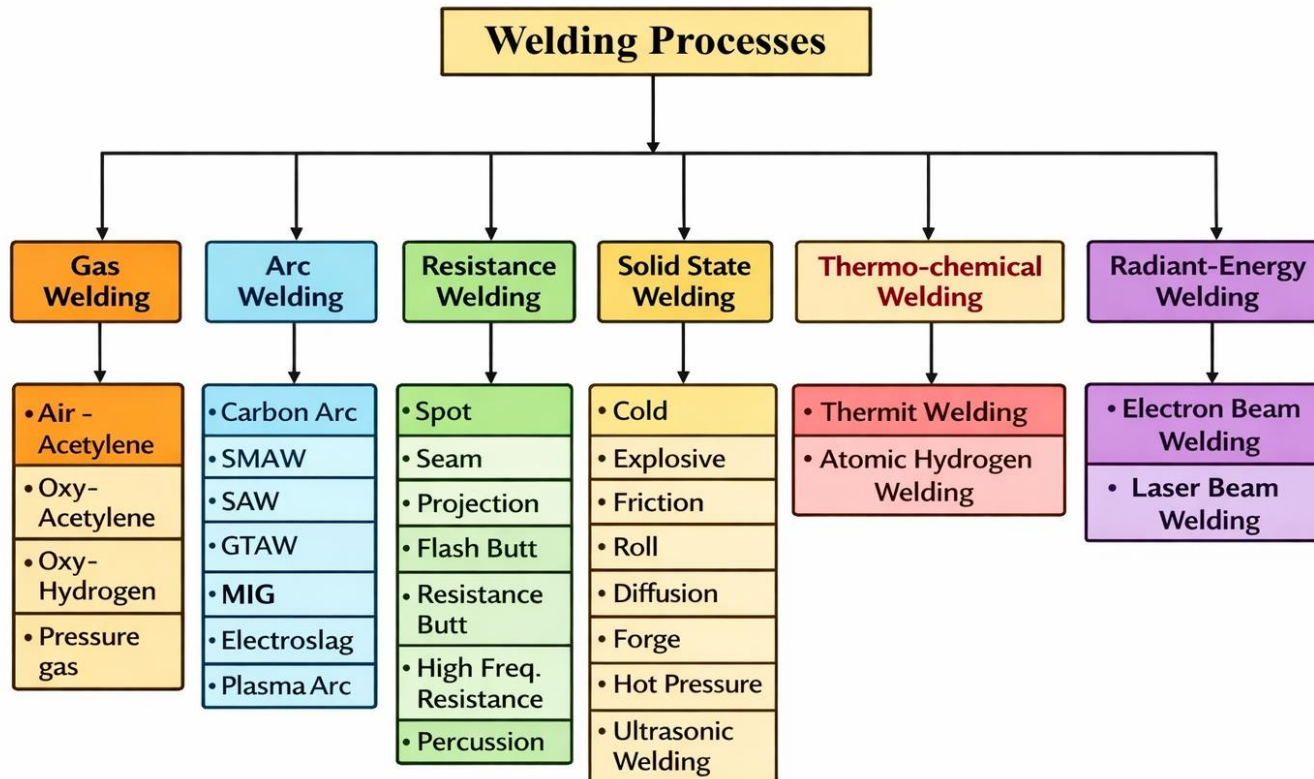


# Types of welding



# Types of welding

## Classification of Welding Processes



**Autogenous:** Solid State    **Homogenous:** Liquid State    **Heterogeneous:** Mixed State

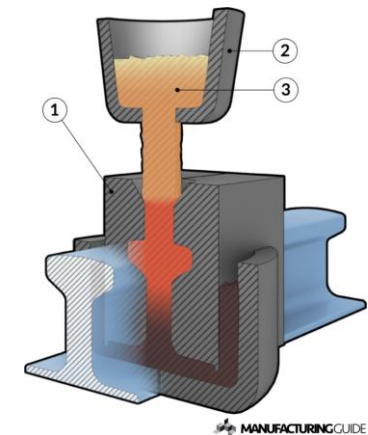
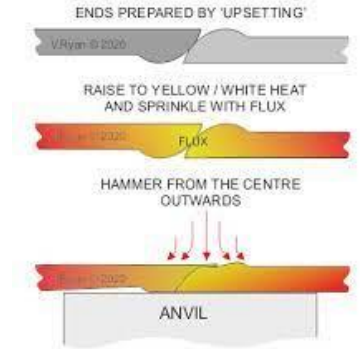
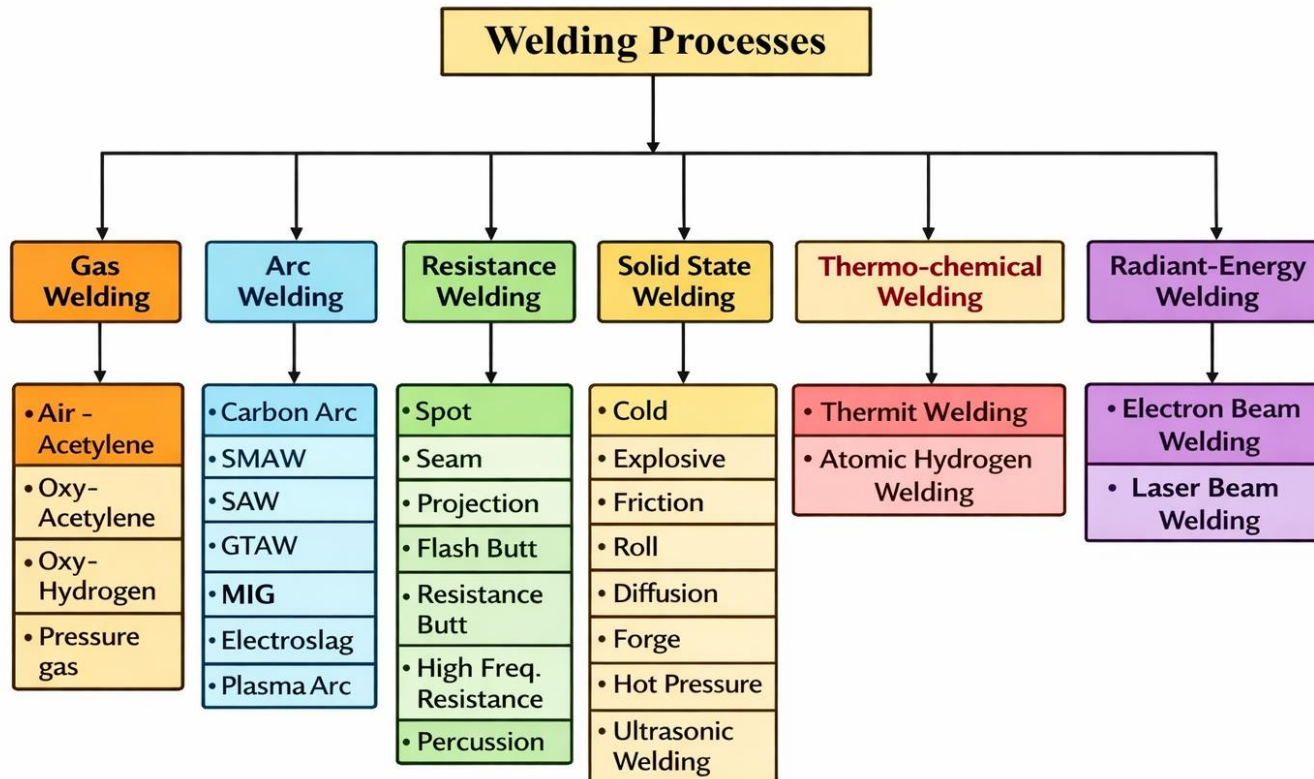


# Types of welding



# Types of welding

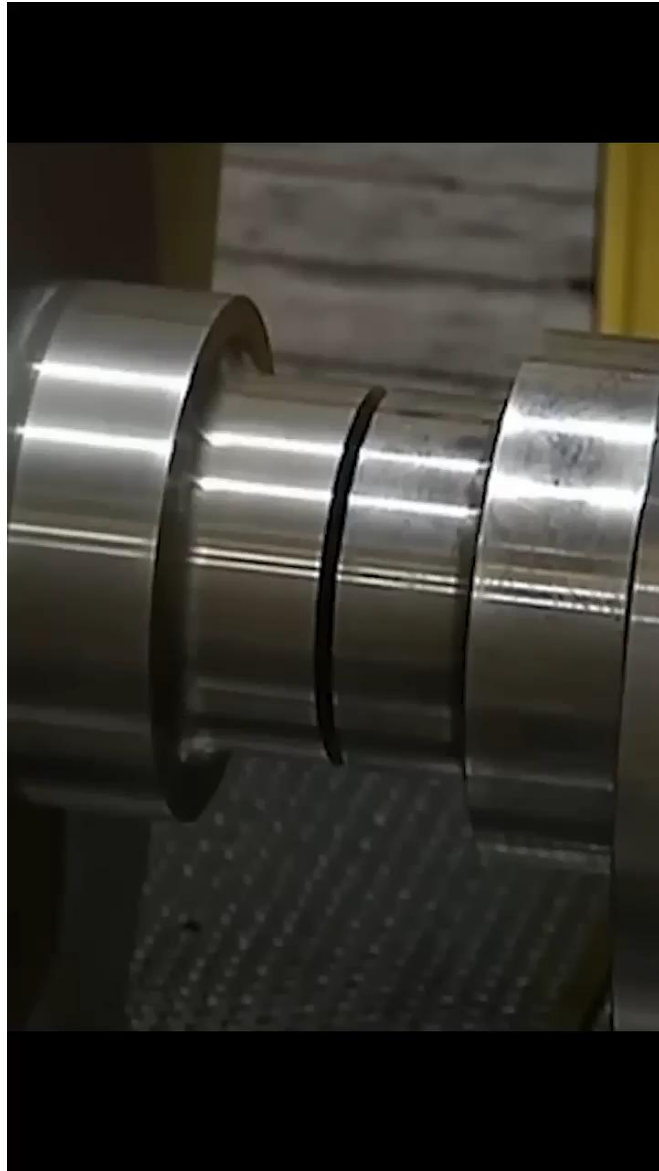
## Classification of Welding Processes



**Autogenous:** Solid State    **Homogenous:** Liquid State    **Heterogeneous:** Mixed State

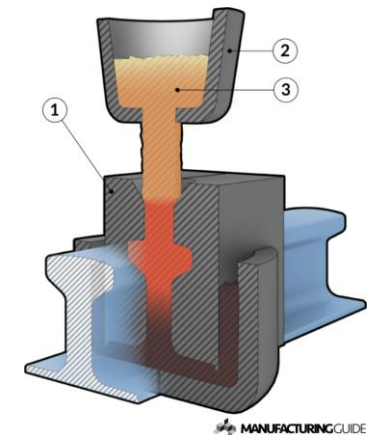
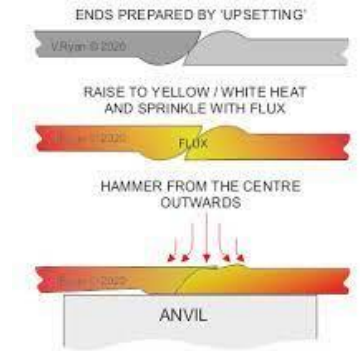
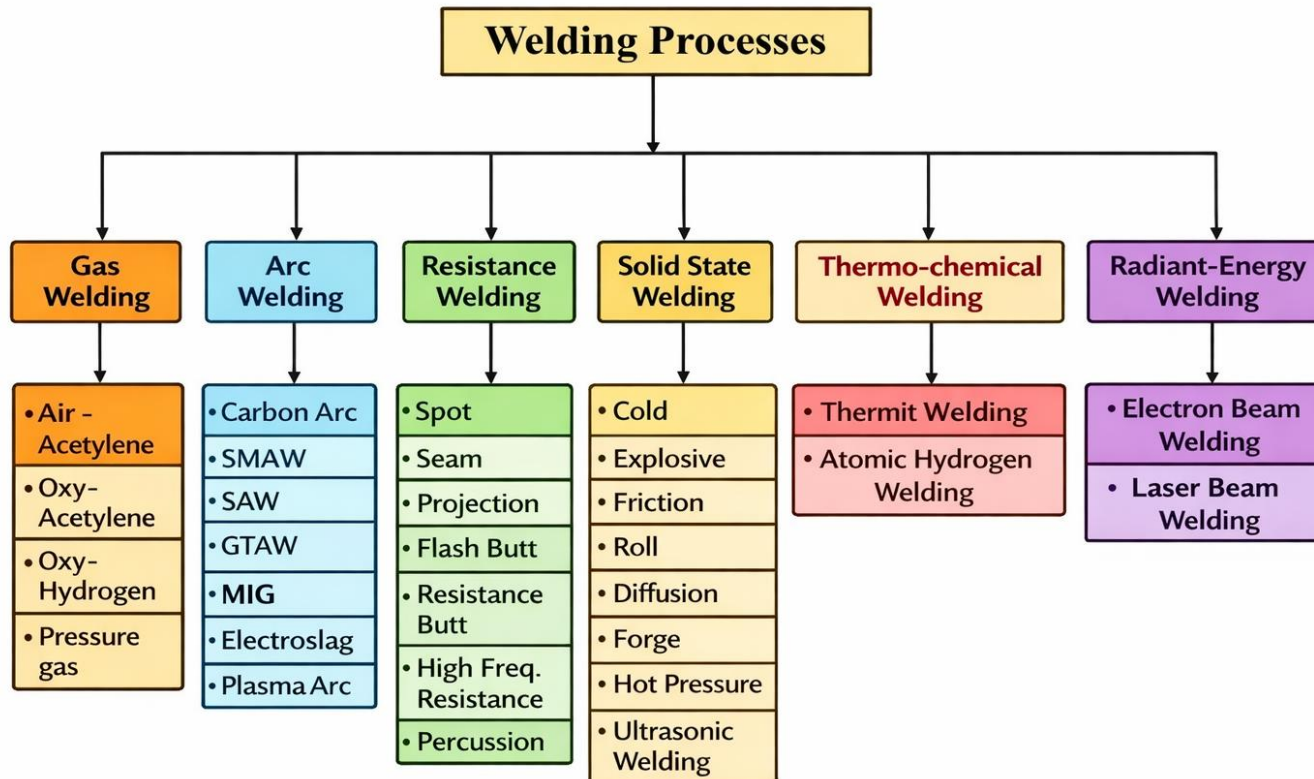
# Types of welding

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# Types of welding

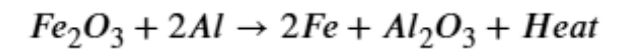
## Classification of Welding Processes



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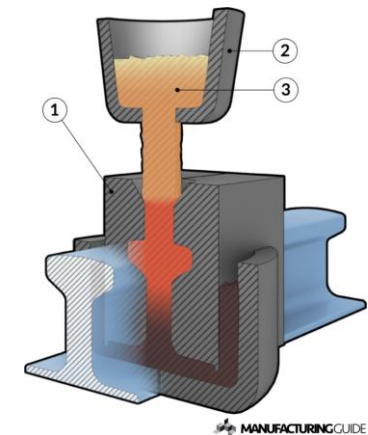
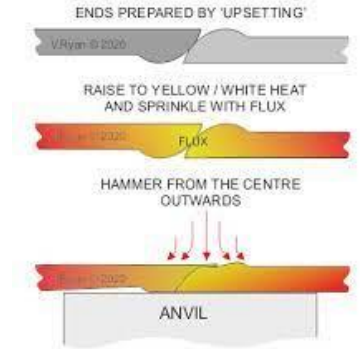
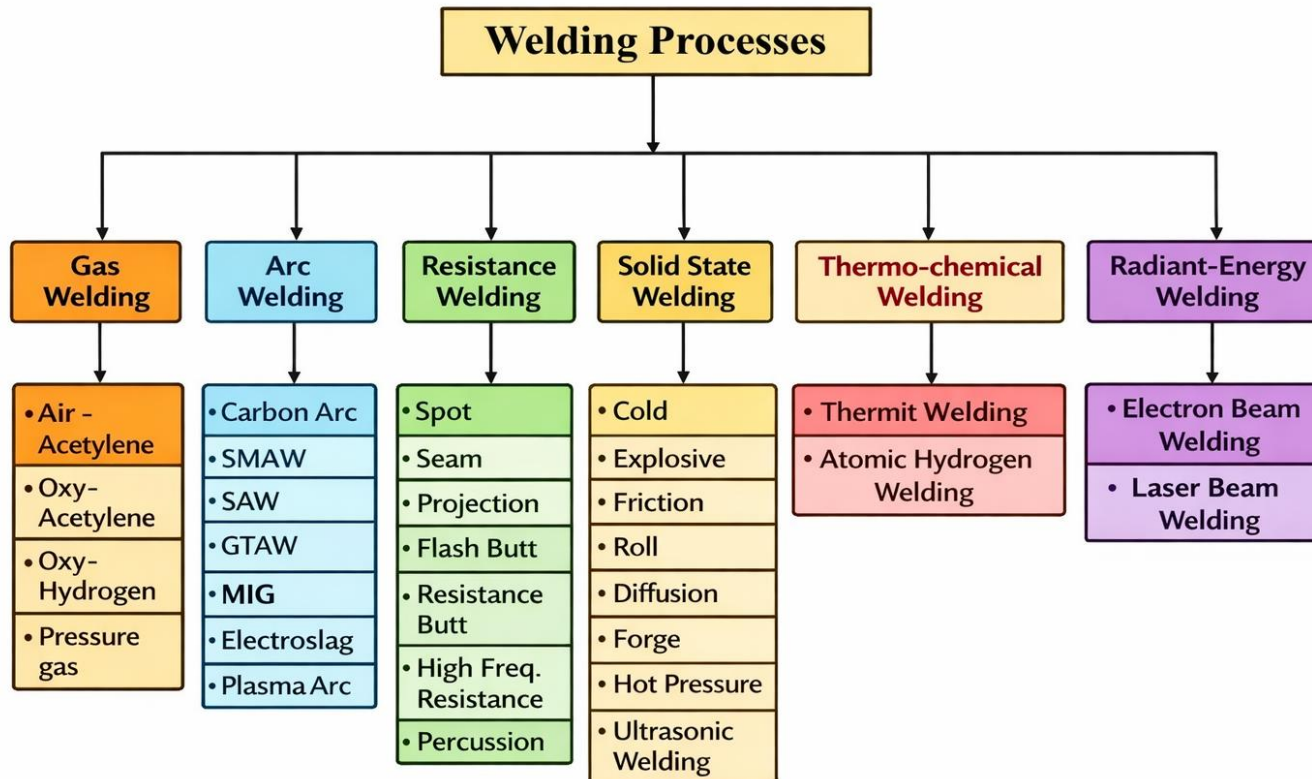


# Types of welding



# Types of welding

## Classification of Welding Processes



**Autogenous:** Solid State    **Homogenous:** Liquid State    **Heterogeneous:** Mixed State

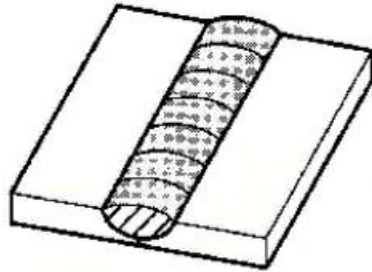
# Types of welding

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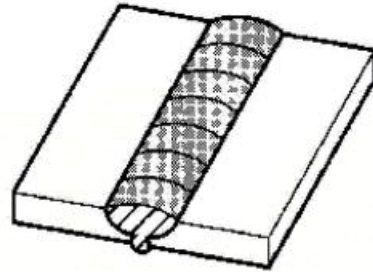


# Types of Welds and Joints

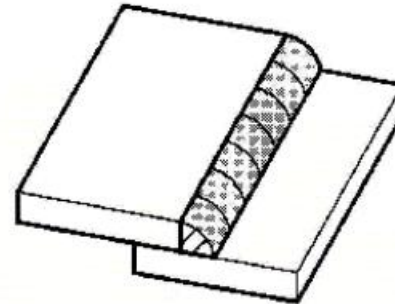
## Types of Fusion welds



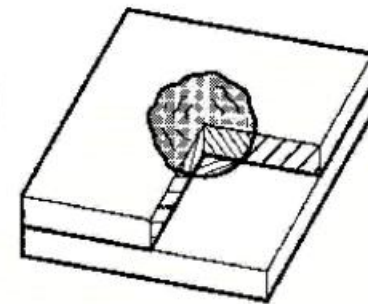
Bead weld  
(or surfacing weld)



Groove weld

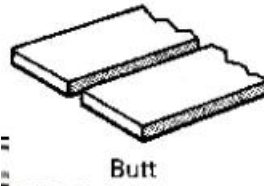


Fillet weld

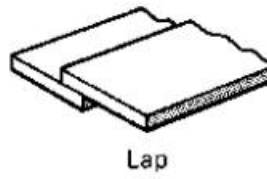


Plug weld

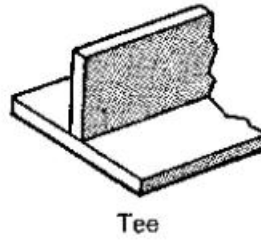
## Types of joints



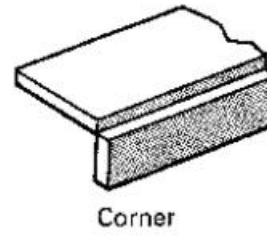
Butt



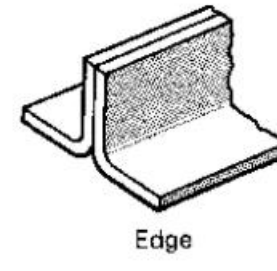
Lap



Tee



Corner



Edge



# Examples of Welded Joints

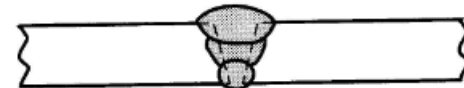
Butt joints



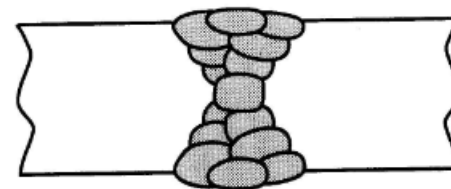
Single-pass square-groove butt joint



Double-pass square-groove butt joint

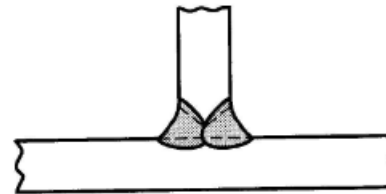


Single V-groove butt joint

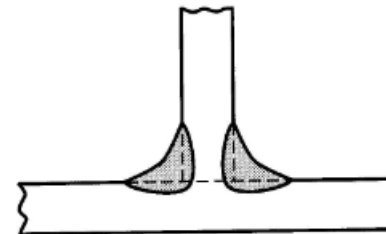


Double V-groove butt joint

T joints

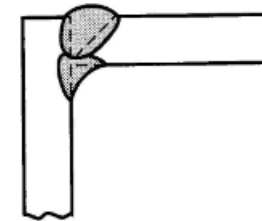


Double bevel-groove T joint

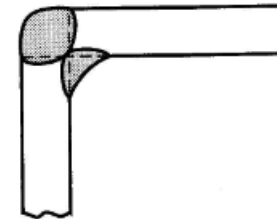


Two-fillet T joint

Corner joints



Single bevel-groove corner joint

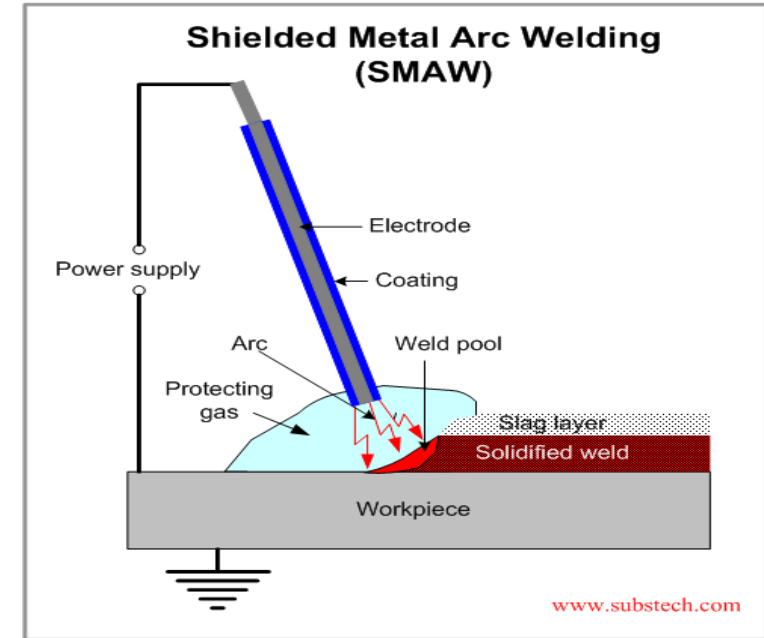


Two fillet corner joint

# Arc welding

## Arc Welding

- Method of fusion welding in which the metals at the joint is heated to a molten state by an electric arc
- Arc column is generated between an anode (electrode) and the cathode (metal to be joined)
- Electrical energy is converted to heat energy in the arc
- Temperature of the arc is about **6000°C** to **7000°C**



# Types of welding

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## Electrodes

**Non- consumable electrodes:** carbon, graphite or tungsten (they are not consumed during welding operation)

**Consumable electrodes:** steel (consumed during welding operation), electrode may be bare (uncoated) or flux-coated

In shielded metal arc welding, the protection to the weld pool is provided by a covering of:

- a) slag formed over the surface of the weld pool/ metal*
- b) inactive gases generated through the thermal decomposition of flux/ coating materials on the electrode*

# Types of ARC welding



## Different Types of Arc Welding

### SMAW

Shielded Metal Arc Welding



Shielded Metal Arc Welding

### GMAW

Gas Metal Arc Welding (MIG)



Gas Metal Arc Welding (MIG)

### FCAW

Flux-Cored Arc Welding



Flux-Cored Arc Welding

### GTAW

Gas Tungsten Arc Welding (TIG)



Gas Tungsten Arc Welding

### SAW

Submerged Arc Welding



Granular Flux

### PAW

Plasma Arc Welding



Plasma Arc Welding

# Types of ARC welding

## Manual metal arc welding

The electrode is consumed in the arc & provides the filler metal for the joint. The extremely high arc temperature of over 5000 °C permits it to supply a large amount of heat to a small area. Hence, the melting in the base metal is restricted to a narrow zone. Among arc processes, manual metal arc is the most common, versatile, and inexpensive, accounting for 60% of total welding in advanced countries and over 90% in India. It is a manual process and hence depends on the welder's skill and experience. Flux-coated electrode having a core of solid wire (diameter 6.3-1.6 mm, length 450-250mm). It needs a power source, either a transformer supplying AC or a generator/ rectifier supplying DC. The process is suitable for the entire range of plate thickness, and for almost all commercial metals and alloys. It can be used in all welding positions.

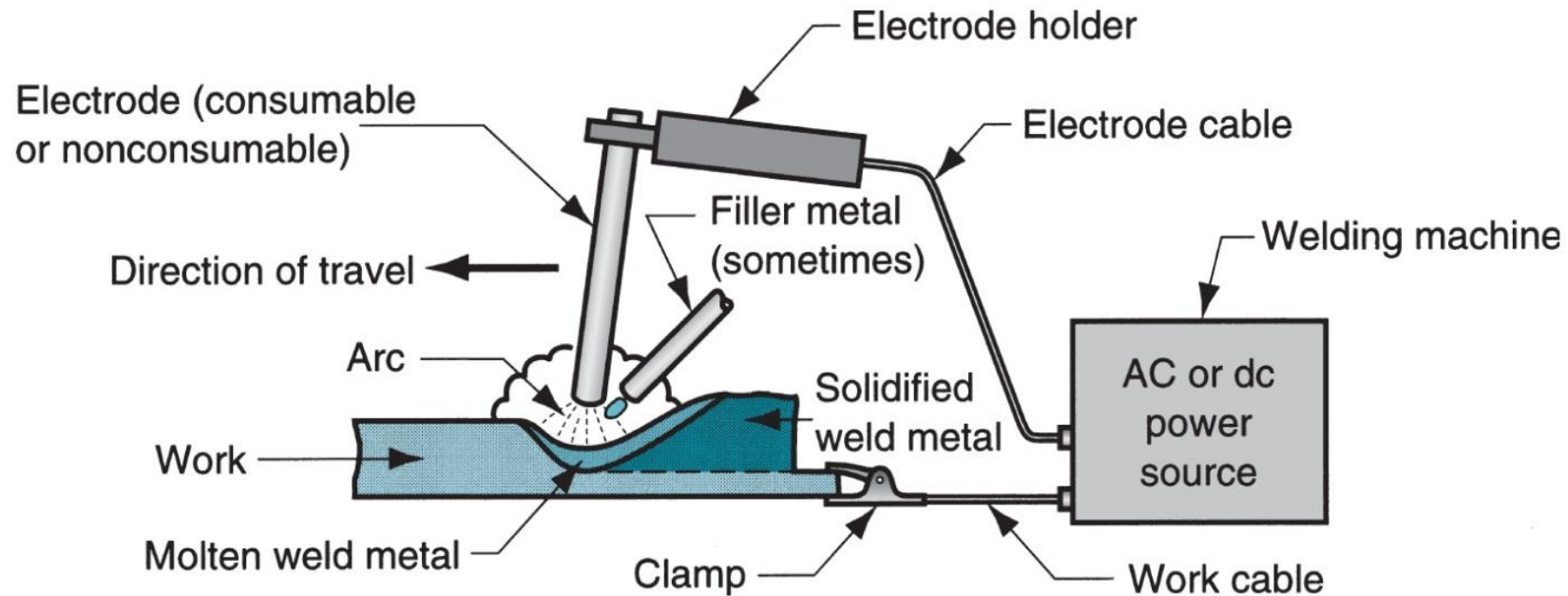


# Manual metal arc welding

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# Manual metal arc welding

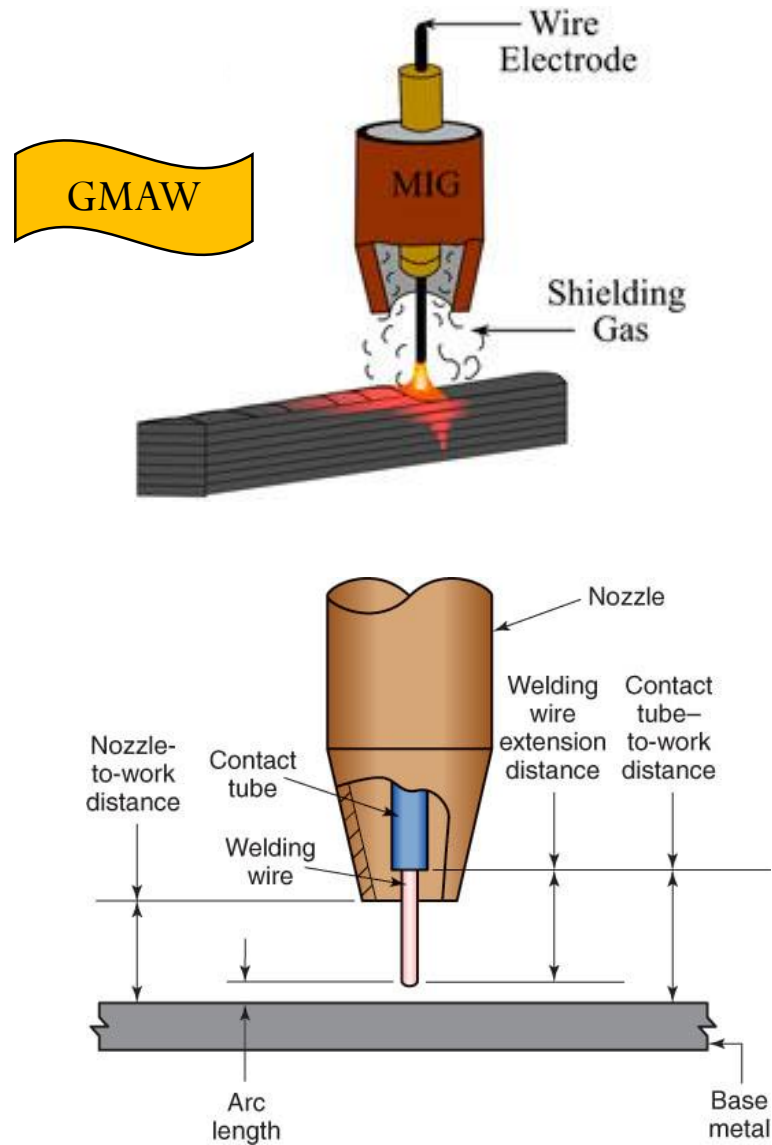


# Gas Metal Arc Welding (GMAW / MIG)

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**Gas Metal Arc Welding (GMAW)**, also known as MIG welding, employs a continuously fed solid wire electrode that acts as both the filler metal and the current carrier. An electric arc forms between the wire and the workpiece, generating sufficient heat to melt the metal, while an externally supplied shielding gas such as argon, CO<sub>2</sub>, or their mixtures protects the weld zone. GMAW is extensively used in automobile manufacturing, sheet metal fabrication, and pressure vessel production because of its high welding speed and clean weld appearance. It is suitable for materials with thicknesses ranging from approximately **1 mm to 25 mm**. The arc temperature is typically around **6,500 to 7,500 °C**, and the process generally operates using a **constant voltage (CV) DC power source**, usually with direct current electrode positive (DCEP).

# Gas Metal Arc Welding (GMAW / MIG)



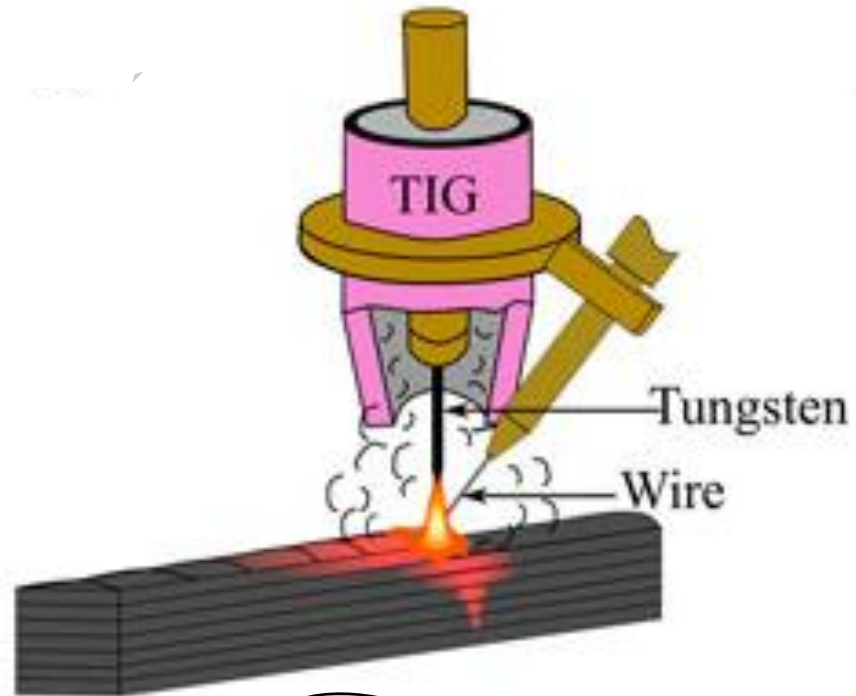
# Gas Tungsten Arc Welding (GTAW/TIG)

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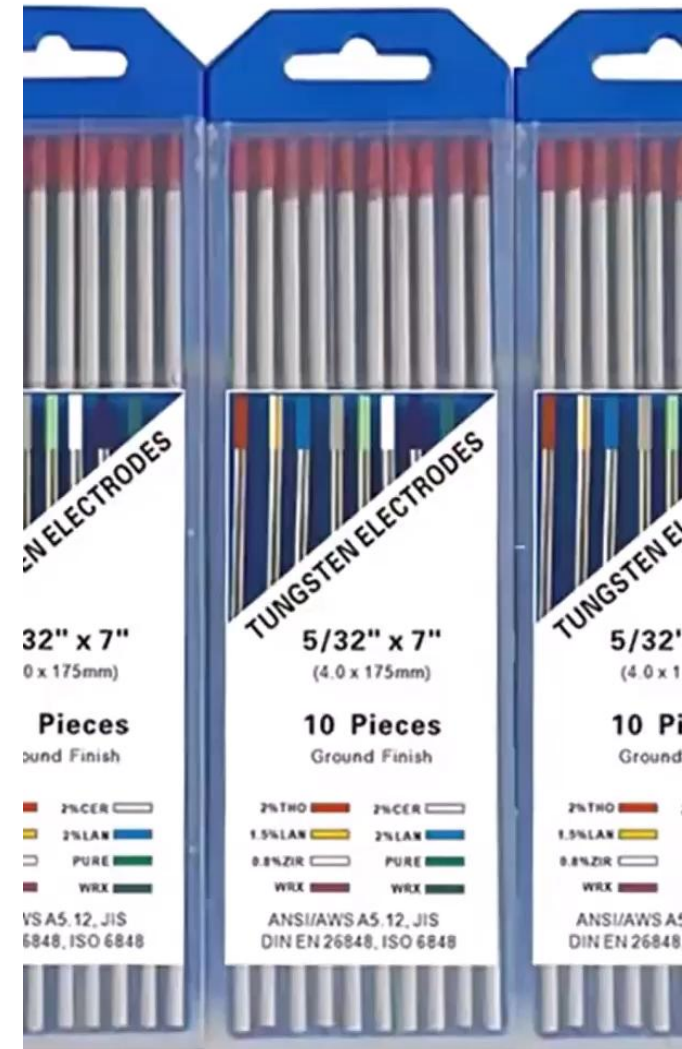
**Gas Tungsten Arc Welding (GTAW)**, or TIG welding, is a high-precision welding process in which the arc is formed between a non-consumable tungsten electrode and the workpiece. The base metal is melted by the arc heat, and filler metal is added separately when required. An inert shielding gas, usually argon or helium, protects the weld area from oxidation. GTAW is widely used in aerospace, nuclear, and chemical industries and is particularly suitable for welding stainless steel, aluminum, and other non-ferrous metals. It is best suited for thin sections, typically in the range of **0.5 mm to 6 mm**, though thicker sections up to about 10 mm can be welded using multiple passes. The arc temperature in GTAW can reach approximately **11,000 °C**, and the process operates with a **constant current (CC) power source**, using AC for aluminum and magnesium alloys and DC for steels and other metals.



# Gas Tungsten Arc Welding (GTAW/TIG)



GTAW



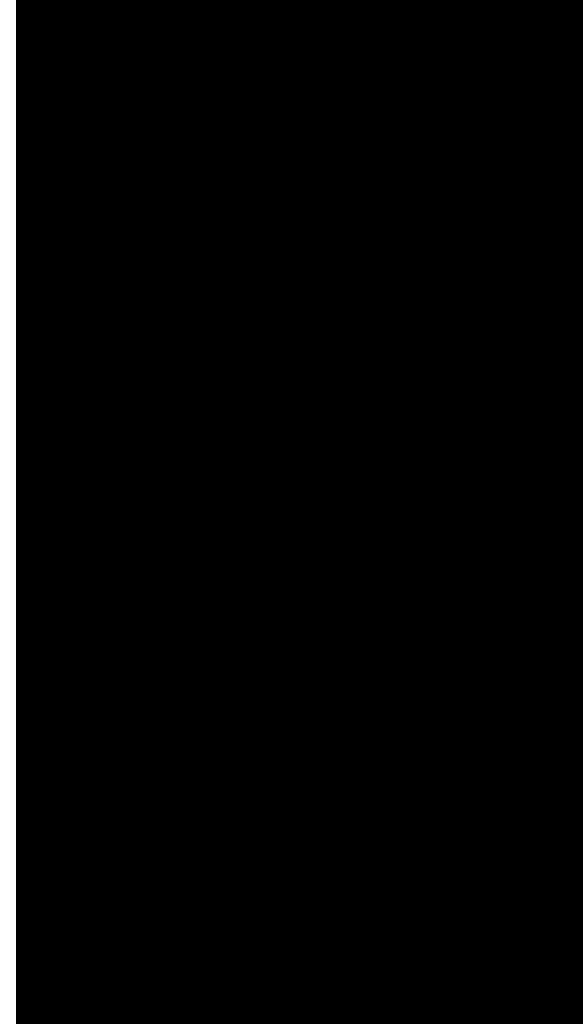
# Submerged Arc Welding (SAW)

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**Submerged Arc Welding (SAW)** is a highly efficient automatic or semi-automatic welding process in which the arc is formed between a continuously fed wire electrode and the workpiece, completely submerged under a blanket of granular flux. The flux prevents spatter, reduces arc radiation, and provides excellent protection to the molten weld pool. SAW is extensively used for welding thick plates in applications such as pressure vessels, large pipes, storage tanks, and ship hulls. It is particularly suitable for welding material thicknesses from **10 mm up to 100 mm**. The arc temperature typically lies between **7,000 and 8,000 °C**, and the process commonly uses a **constant voltage (CV) power source**, operating with either AC or DC, depending on the application.

# Submerged Arc Welding (SAW)

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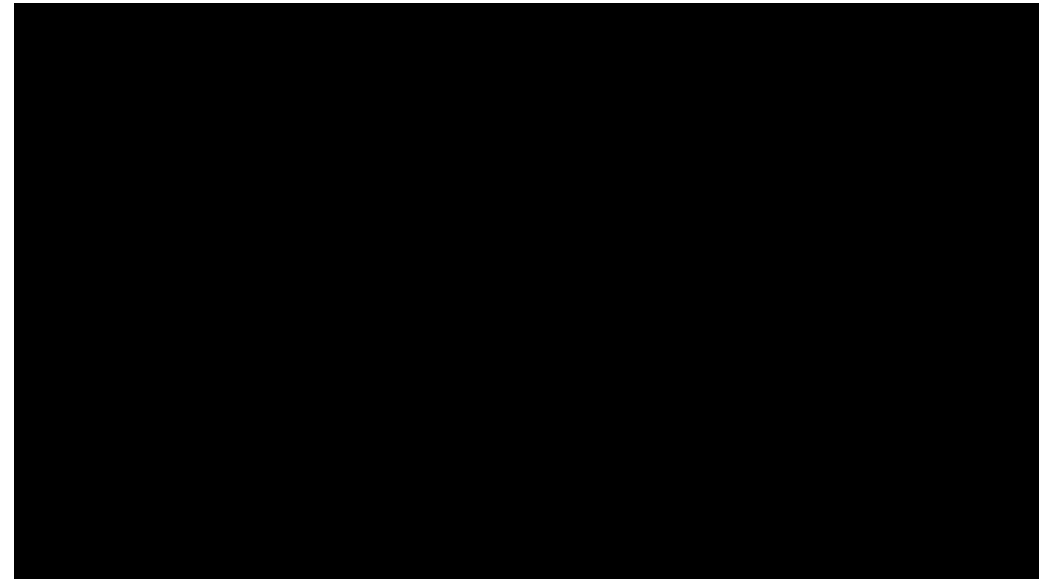
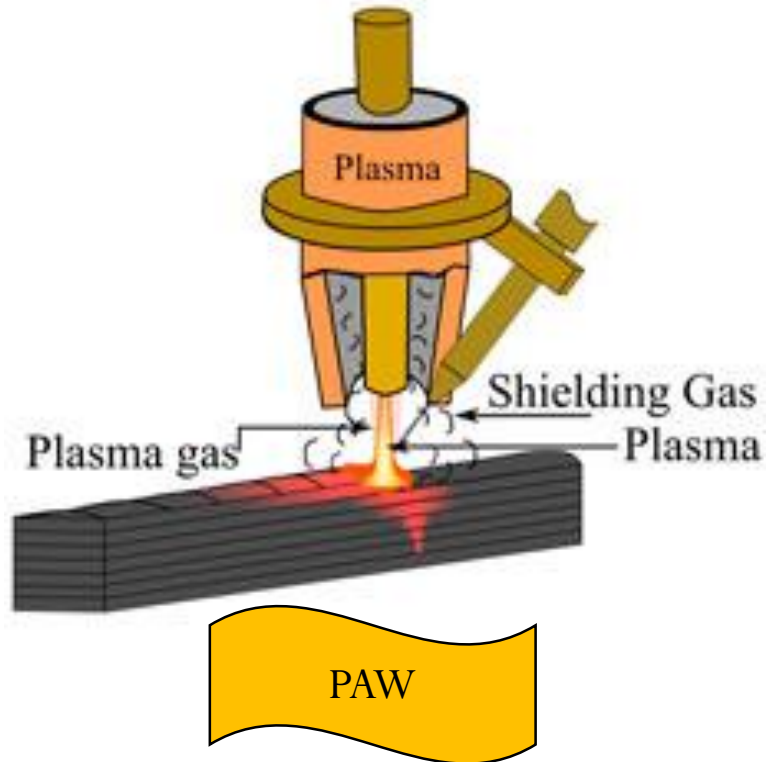
# Plasma Arc Welding (PAW)

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**Plasma Arc Welding (PAW)** is an advanced arc welding process that utilizes a highly constricted plasma arc formed by forcing ionized gas through a narrow nozzle. This produces a very high-energy, concentrated arc capable of precise and controlled welding. PAW is widely applied in aerospace, medical device manufacturing, and other precision industries where high accuracy and repeatability are required. It is suitable for welding extremely thin materials, starting from about **0.1 mm**, and can be used up to approximately **8 mm** thickness. The plasma arc temperature is exceptionally high, typically in the range of **16,000 to 20,000 °C**, and the process operates using a **constant current (CC) DC power source**.

# Plasma Arc Welding (PAW)

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# Arc welding

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## **Advantages**

1. Simple and equipment is portable
2. Almost all metals and alloys can be welded using this technique
3. Good welds can be obtained

## **Disadvantages:**

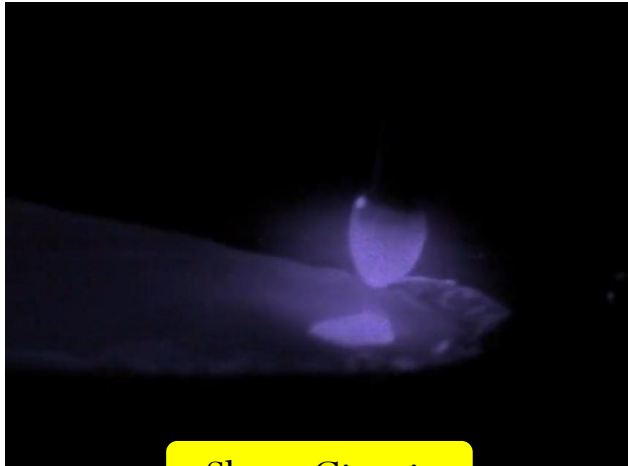
1. Slow process
2. Requires post-cleaning (to remove slag)

## **Applications:**

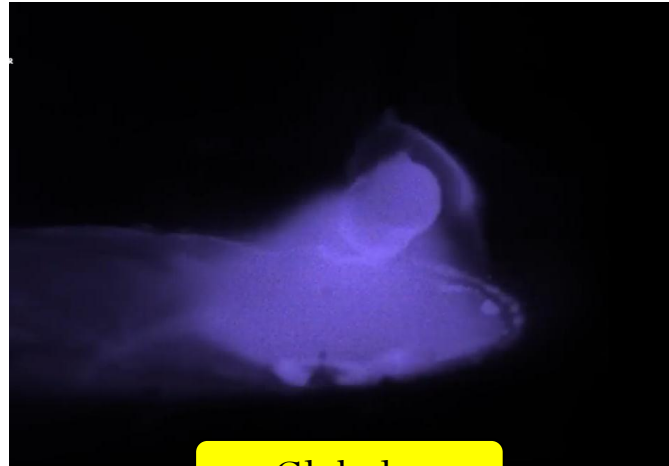
Used to weld tanks, boilers, pressure vessels, shipbuilding, pipe joining etc.

# Metal Transfer Modes

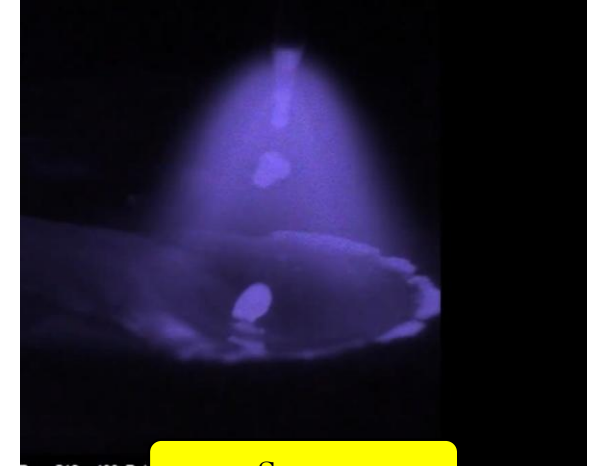
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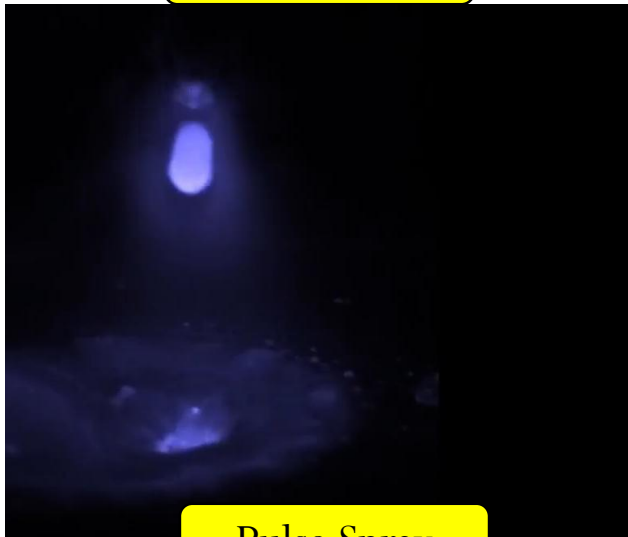
Short Circuit



Globular



Spray



Pulse Spray

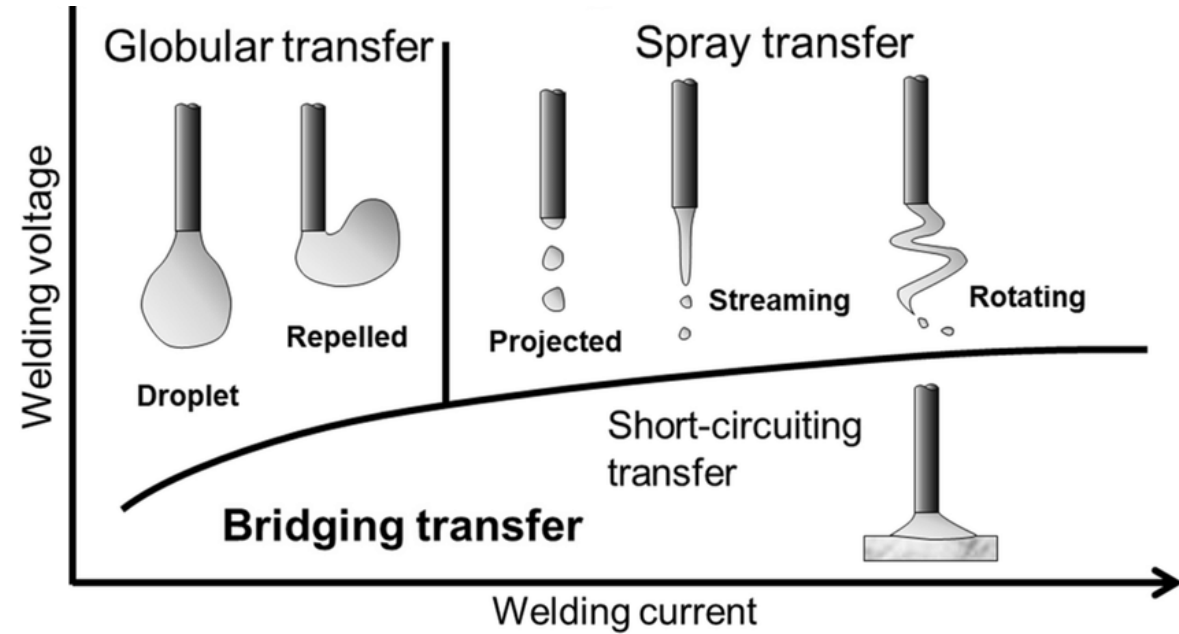
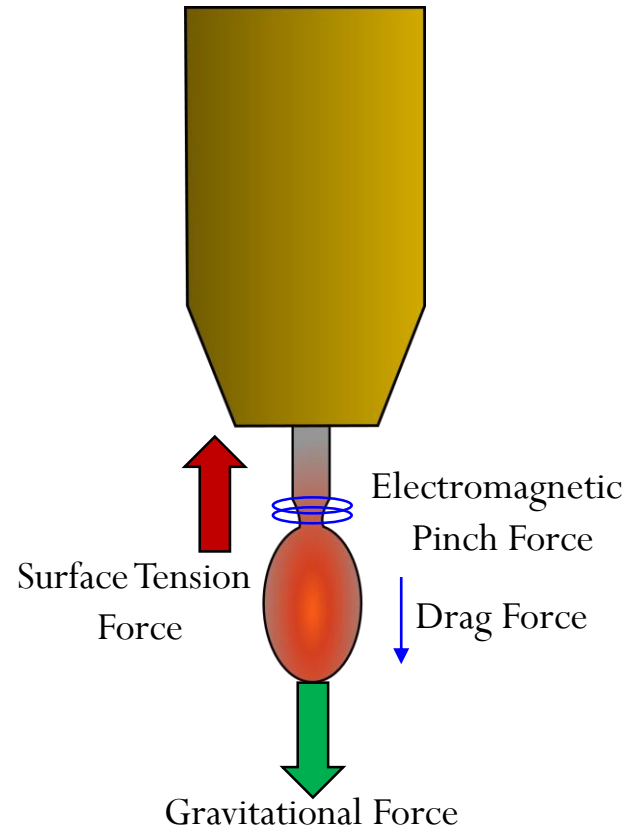


Rotating Spray



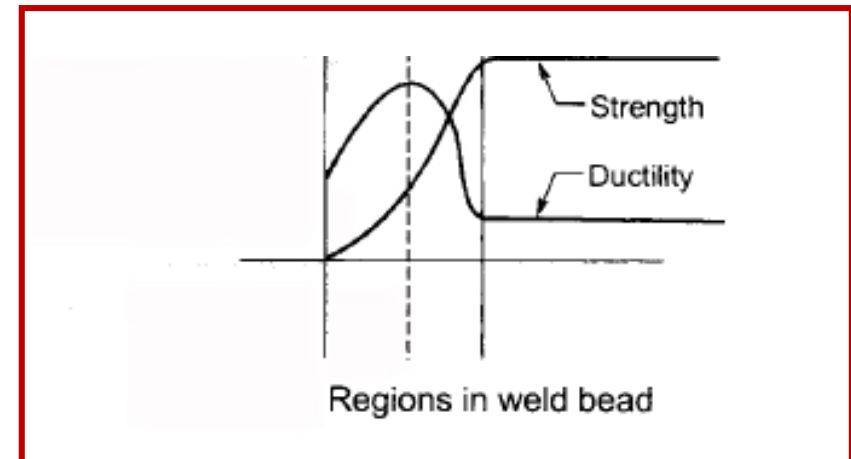
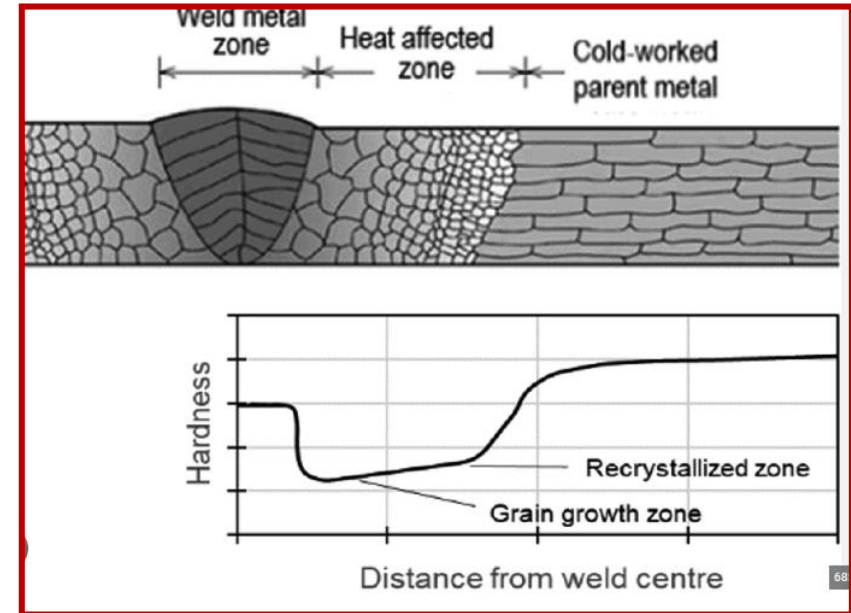
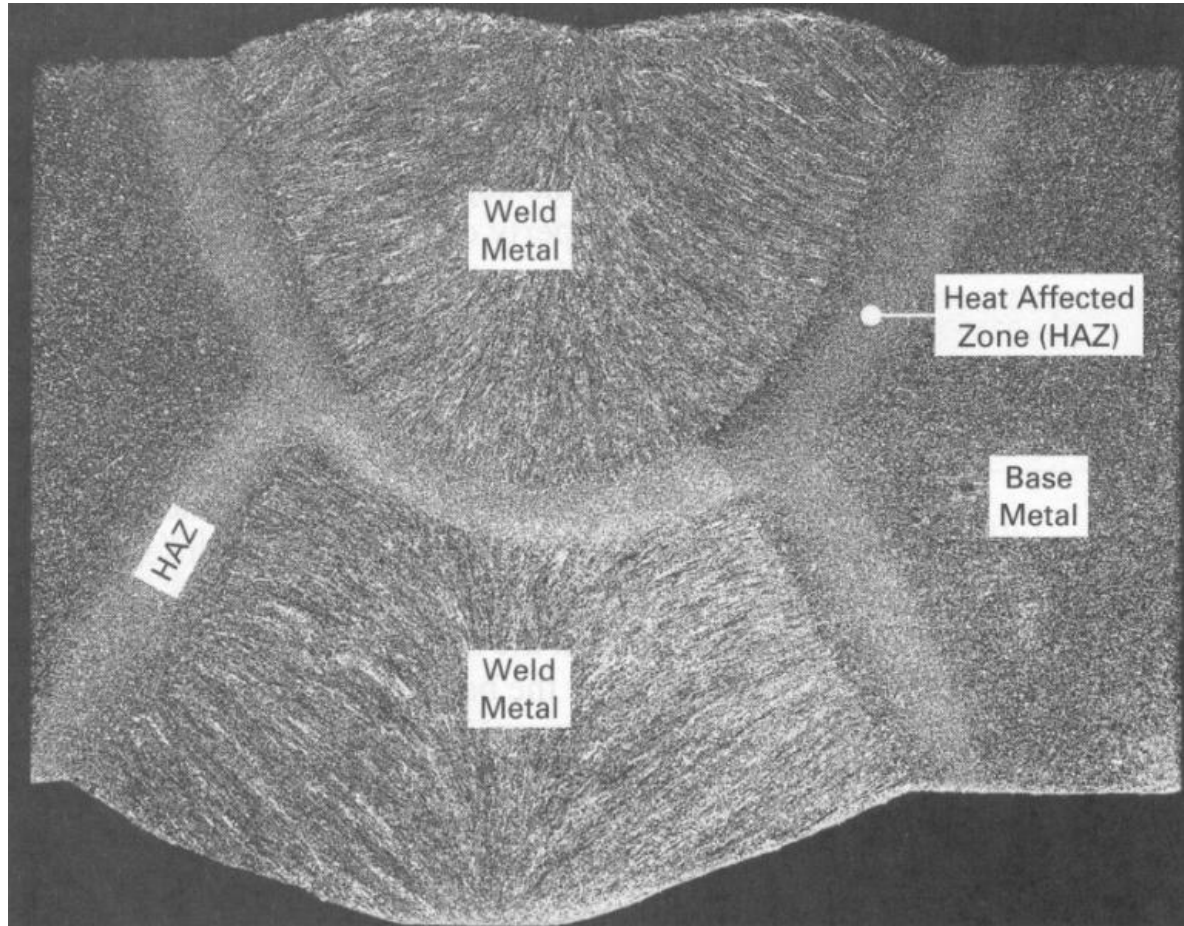
100% Argon

# Metal Transfer Modes



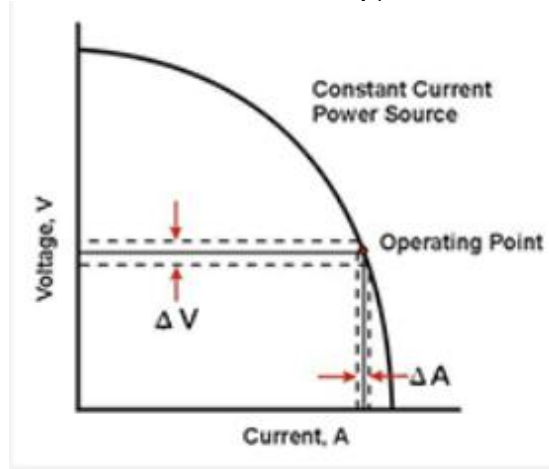
| Metal Transfer Mode | Deposition Rate (kg/hr) |
|---------------------|-------------------------|
| Short circuit       | 0.9 – 2.7               |
| Globular            | 1.8 – 3.2               |
| Spray               | 2.7 – 5.4               |
| Pulse Spray         | 0.9 – 2.7               |
| Rotating Spray      | 6.9 – 13.5              |

# Typical Weld Microstructure

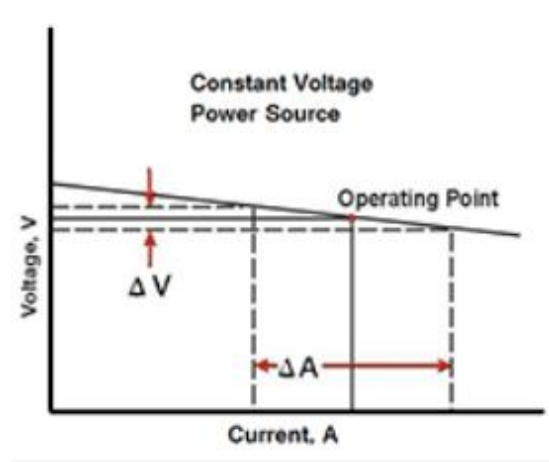


# Power Source

**Constant Current:** Delivers stable amperage, with voltage fluctuating to maintain arc length; ideal for Stick (SMAW) and TIG (GTAW) where manual arc length control varies.



**Constant Voltage:** Provides steady voltage, allowing current to vary with wire feed; perfect for MIG/MAG and Submerged Arc welding.





# Power Source

---

- **Power Source Characteristic:** A DC welding power source typically has a linear V-I curve described by:
  - $V = V_0 - (V_0/I_s) \times I$ .
  - $V_0$  (Open Circuit Voltage) is the voltage with no current.
  - $I_s$  (Short Circuit Current) is the current when voltage is zero.
  - $V$  and  $I$  are the operating voltage and current.
- **Arc Characteristic:** The voltage required to sustain an arc (arc voltage,  $V_{arc}$ ) changes with arc length ( $l$ ):
  - $V_{arc} = A + B \times l$  (e.g.,  $V = 10 + 5l$ ).

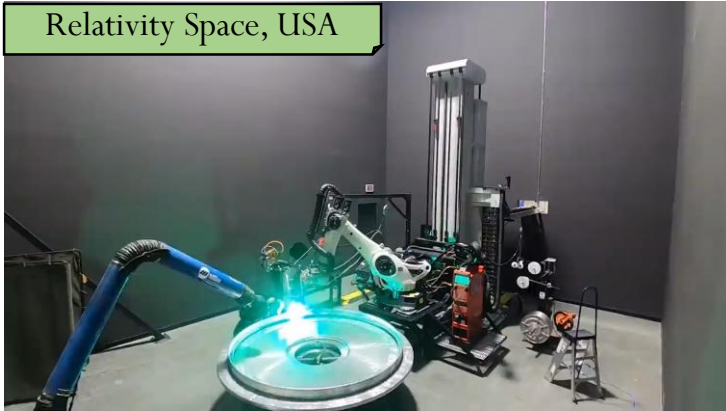
# **Wire Arc Additive Manufacturing**

# Introduction

## Wire Arc Additive Manufacturing

- Uses electric arc as the source of energy to melt the feed stock wire which is available in the form of wire.
- Based on different source of arc generation, WAAM can be classified into: Gas Tungsten Arc Welding (GTAW), Gas Metal Arc Welding (GMAW) and Plasma Arc Welding (PAW).
- High deposition rate (6-8 kg/hr) is obtained with WAAM as compared to traditional additive manufacturing processes

Relativity Space, USA



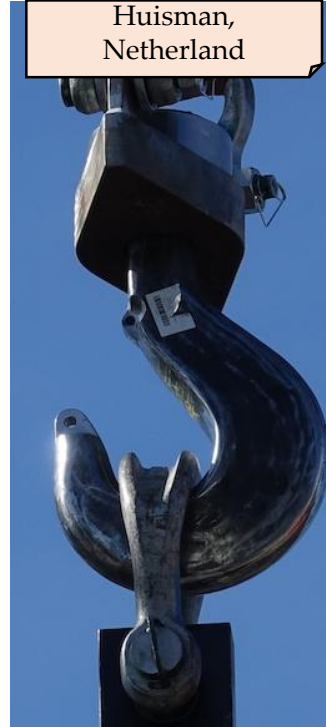
IIT Guwahati, India



MX3D, Netherland

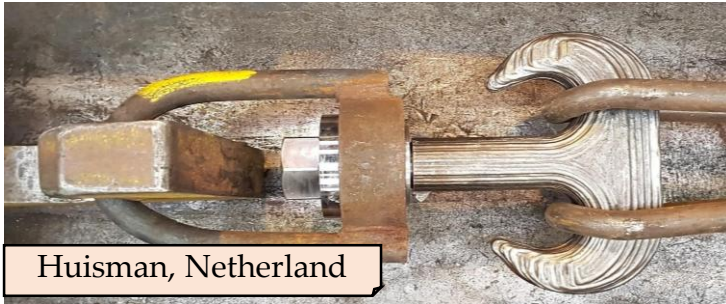


Huisman, Netherland

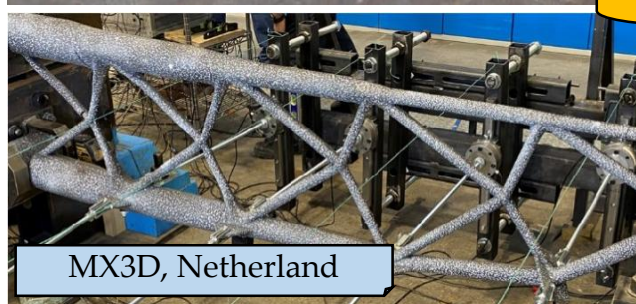


## Application

Huisman, Netherland



MX3D, Netherland



IIT Bombay, India



# Introduction

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**Materials**

**Metal**

**Material Flow**

**Continuous**

**Drop on Demand**

**Machine Kinematics**

**Serial**

**Parallel**

**Hybrid**

**Energy Source**

**Induction**

**Arc**

Wire Arc AM

MIG, TIG, Plasma

**Energy Beam**

Wire Beam AM

Laser, EB

**Support Mechanism**

**Acid bath**

**No Support / Multi- Axis Kinematics**

# Introduction

Materials

Material Flow

Machine Kinematics

Energy Source

Support Mechanism

Metal

Continuous

Drop on Demand

Serial

Parallel

Hybrid

Induction

Arc

Wire Arc AM

Energy Beam

Wire Beam AM

MIG, TIG, Plasma

Laser, EB

Acid bath

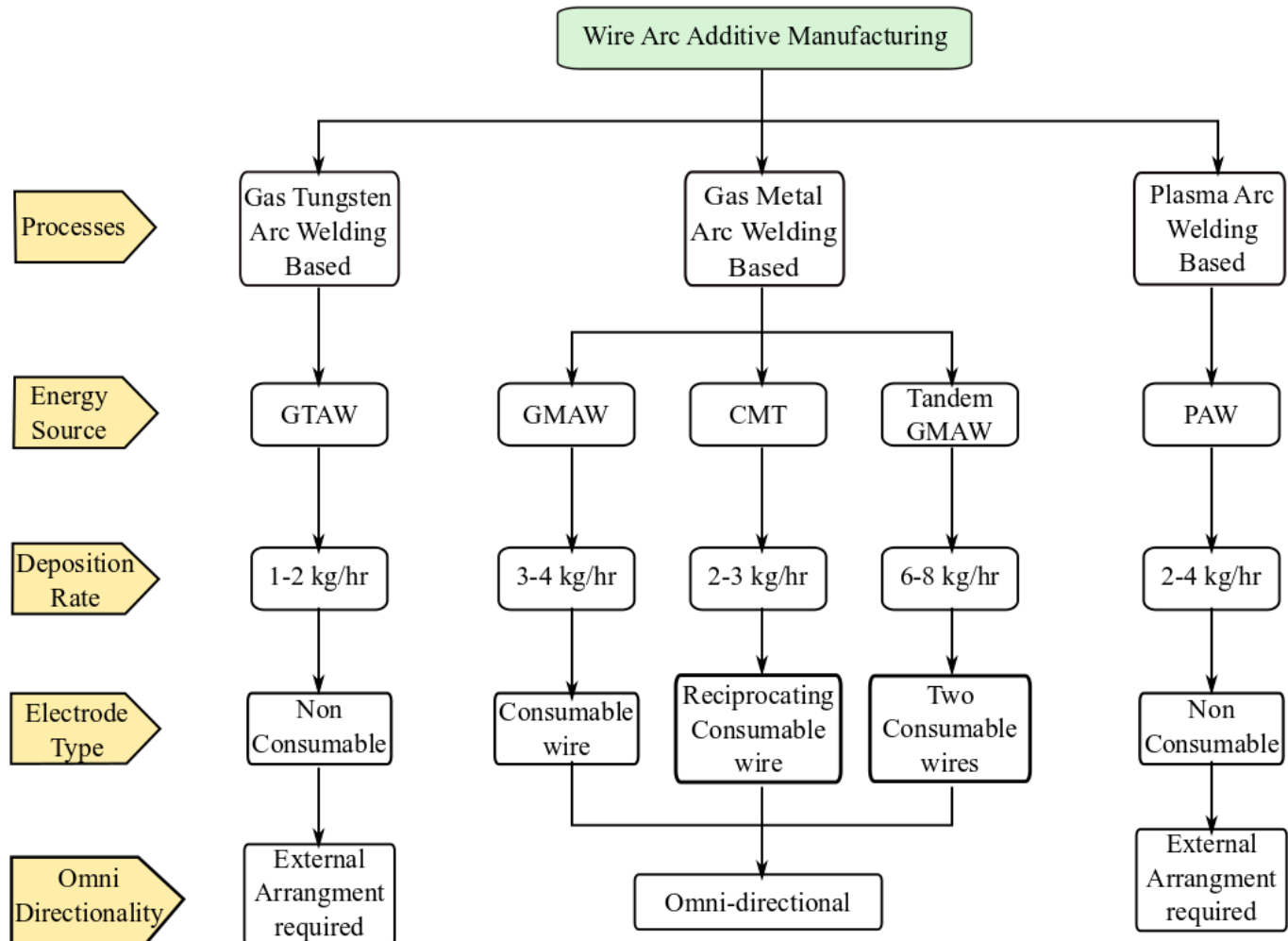
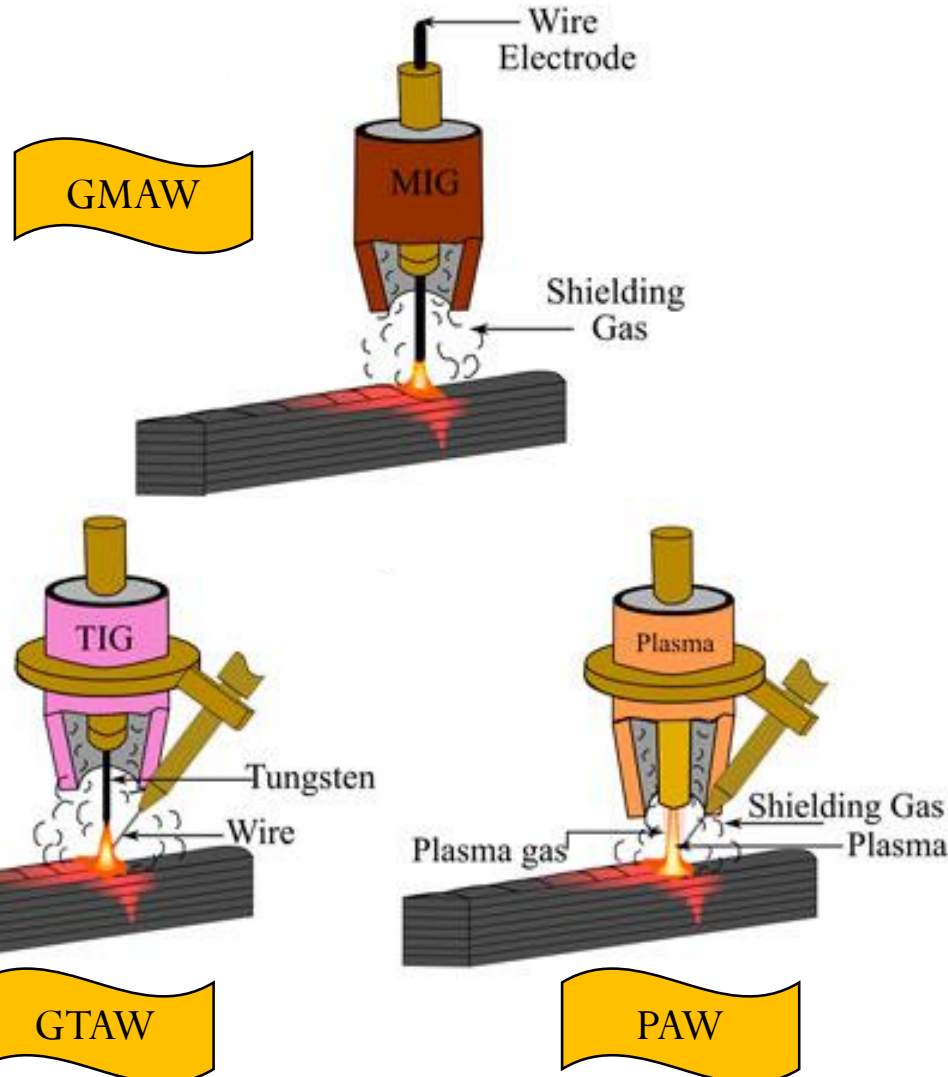
No Support

Multi- Axis Kinematics



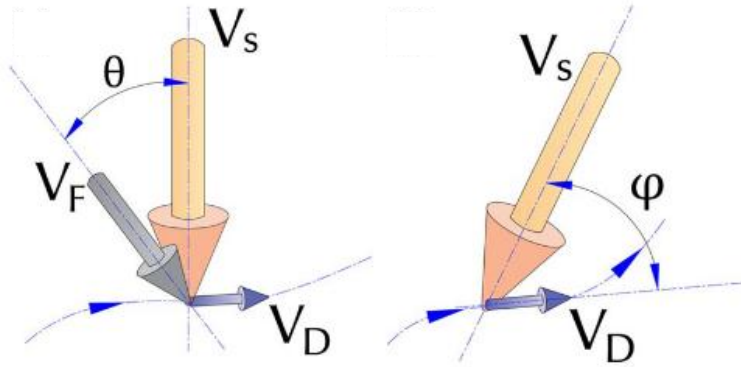
# Wire Arc Additive Manufacturing

## Classifications

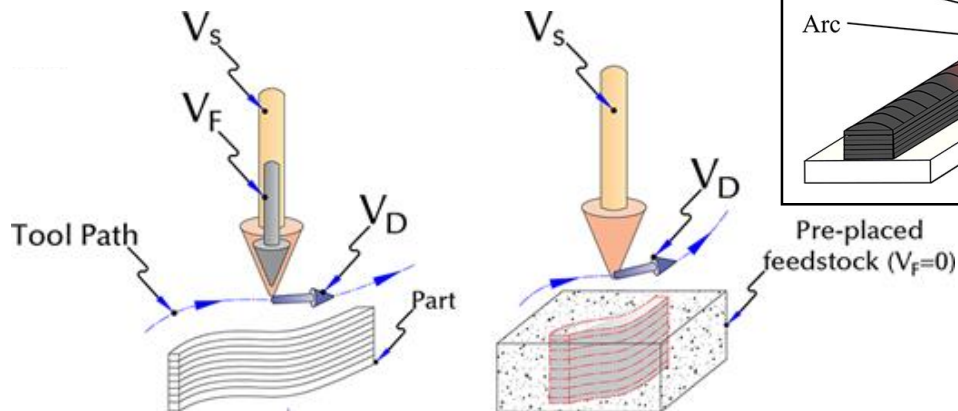
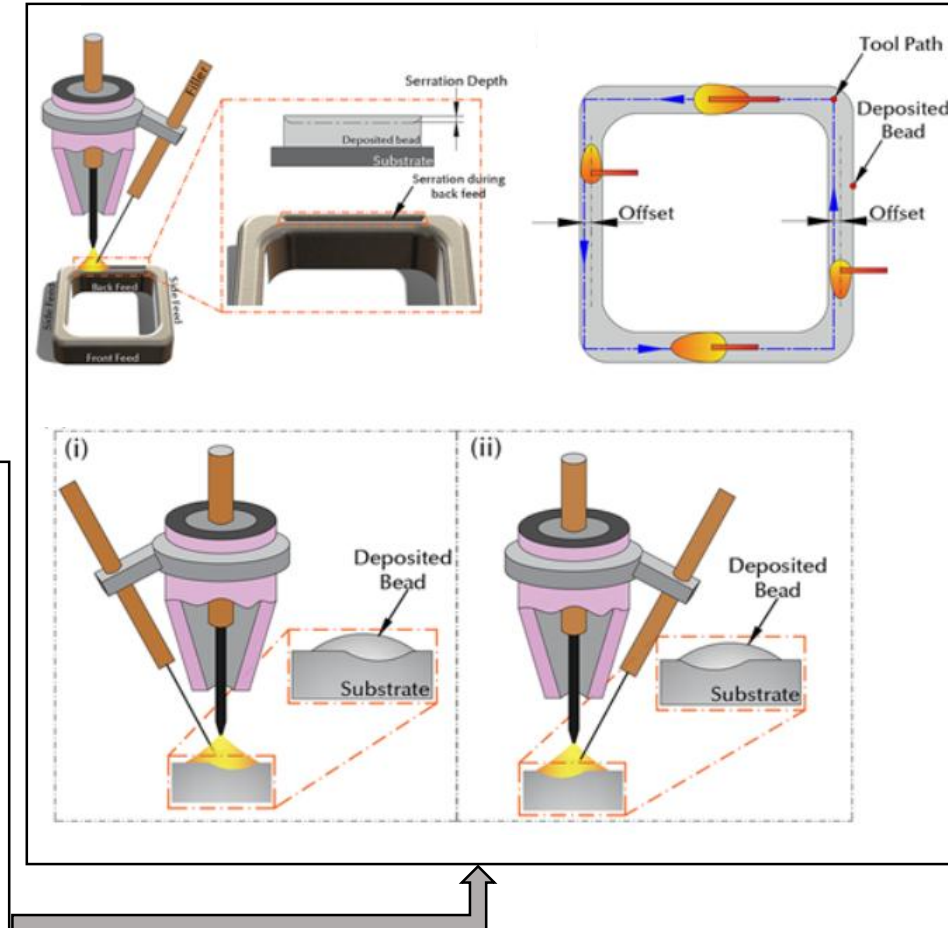
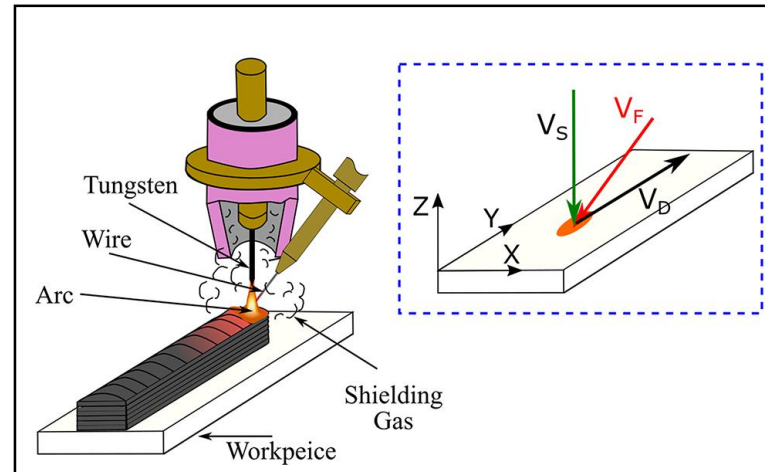
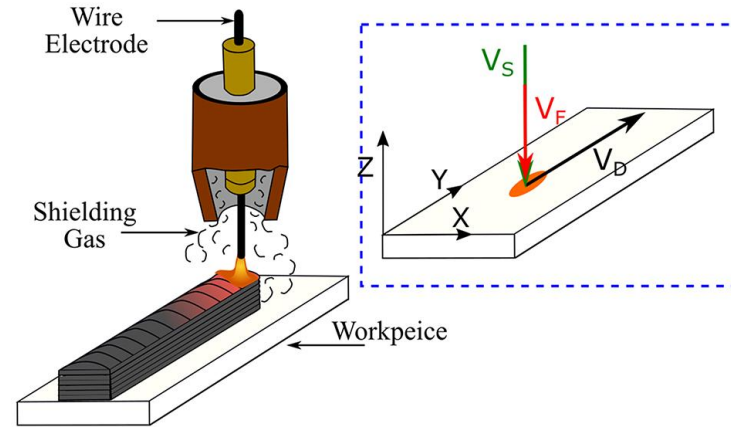


# Wire Arc Additive Manufacturing

## Omnidirectionality



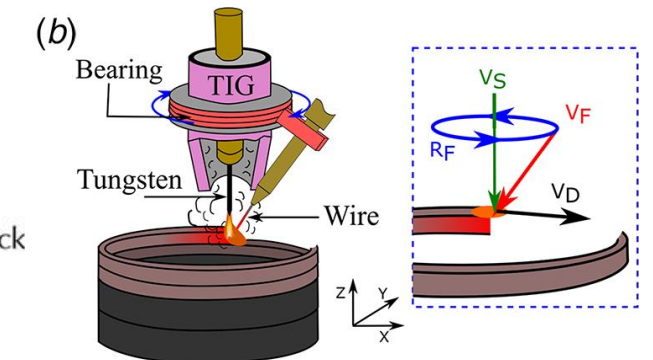
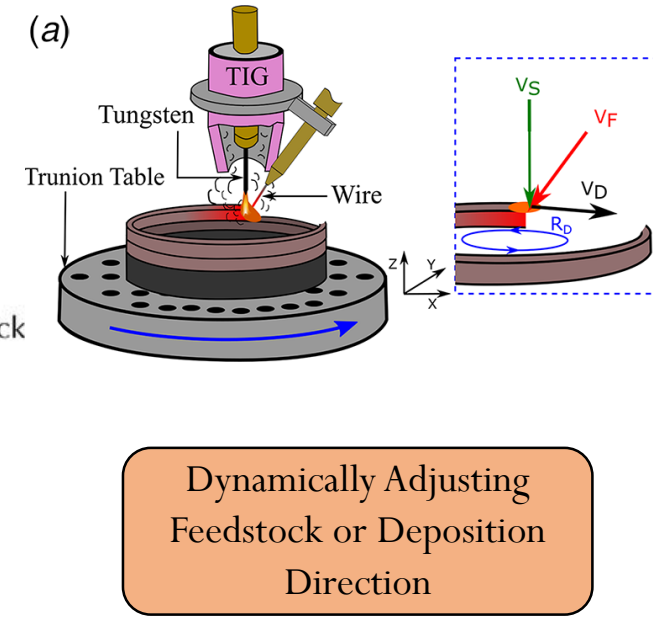
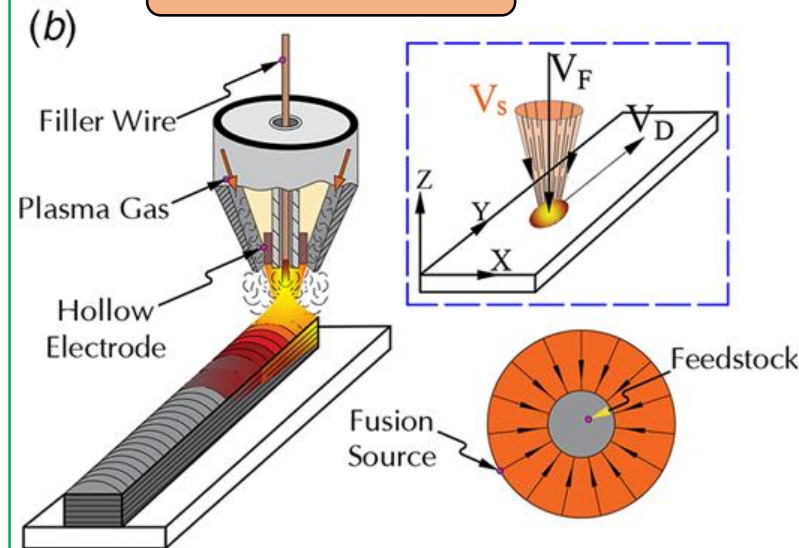
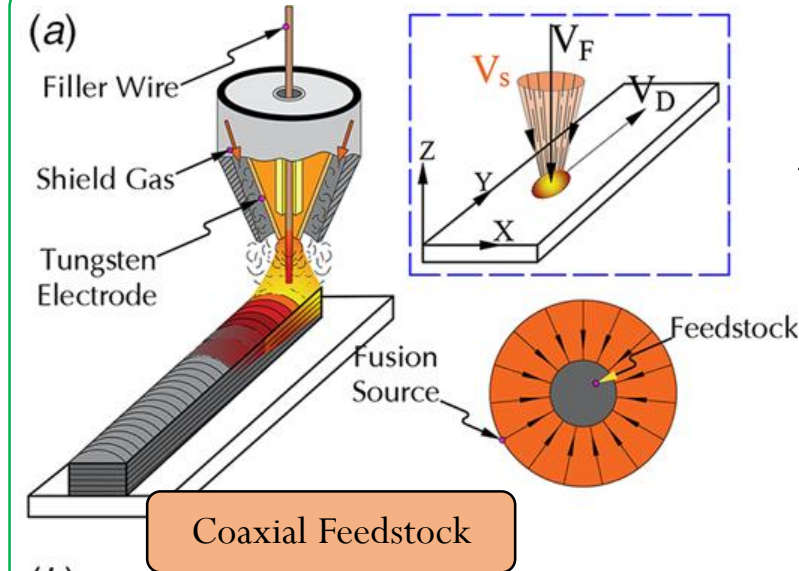
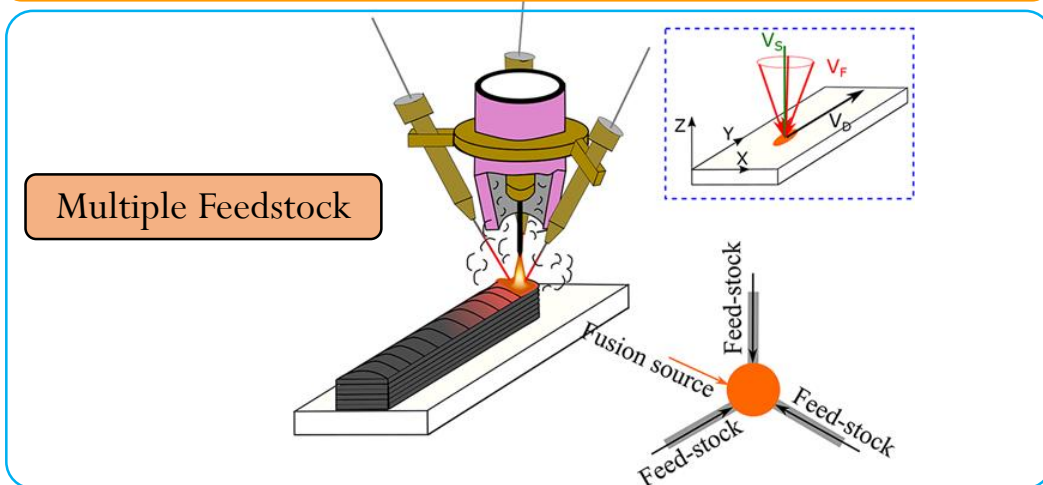
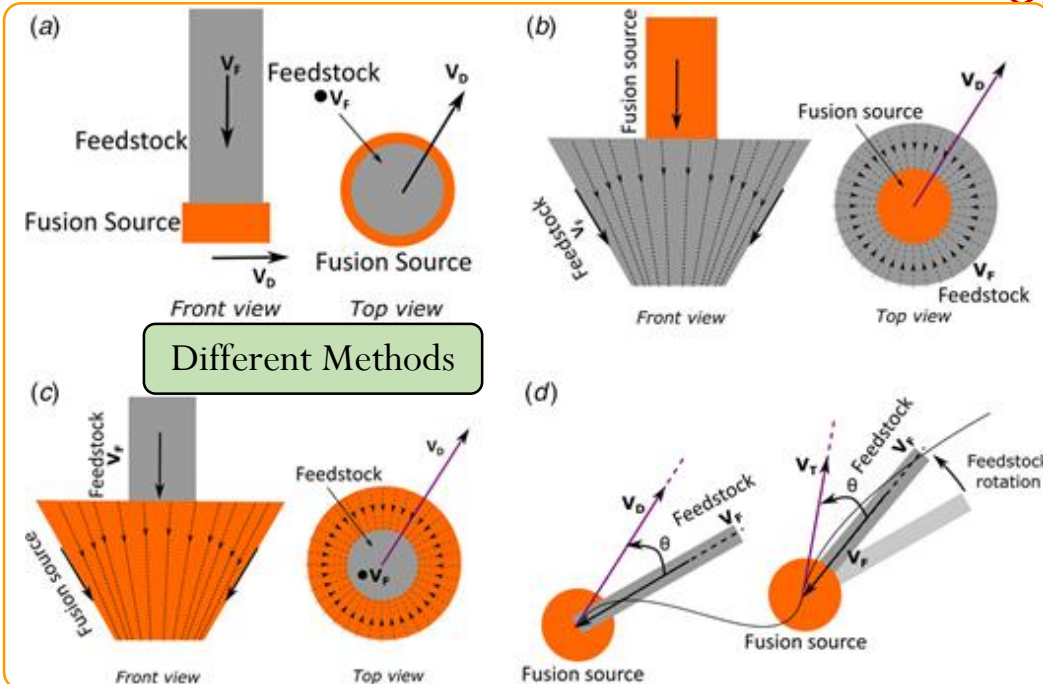
$V_S$  is the Source of Fusion Vector  
 $V_F$  is the Feedstock Vector  
 $V_D$  is the Deposition Vector



$$V_F \parallel V_S ; V_D \perp V_S \text{ and } V_D \perp V_F$$

# Wire Arc Additive Manufacturing

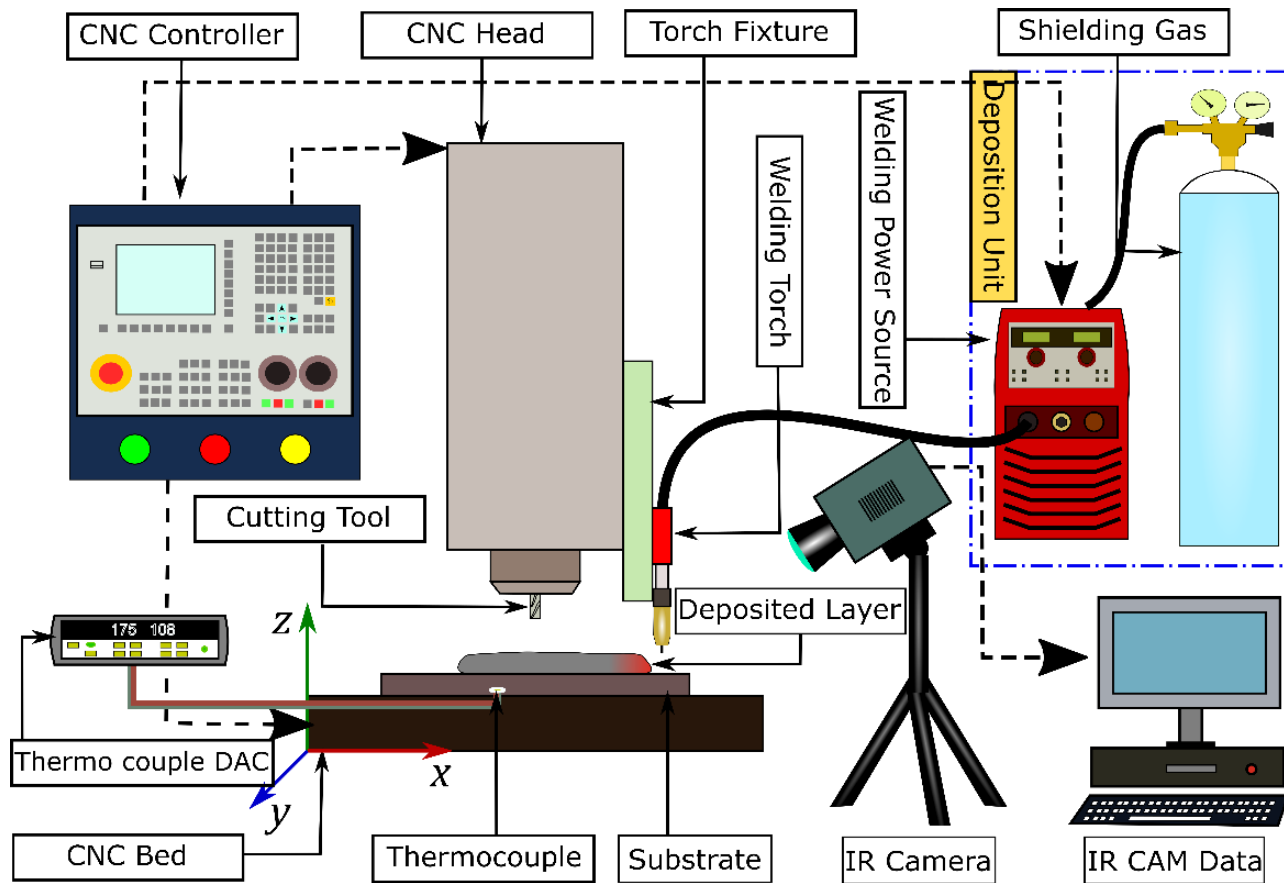
## Omnidirectionality



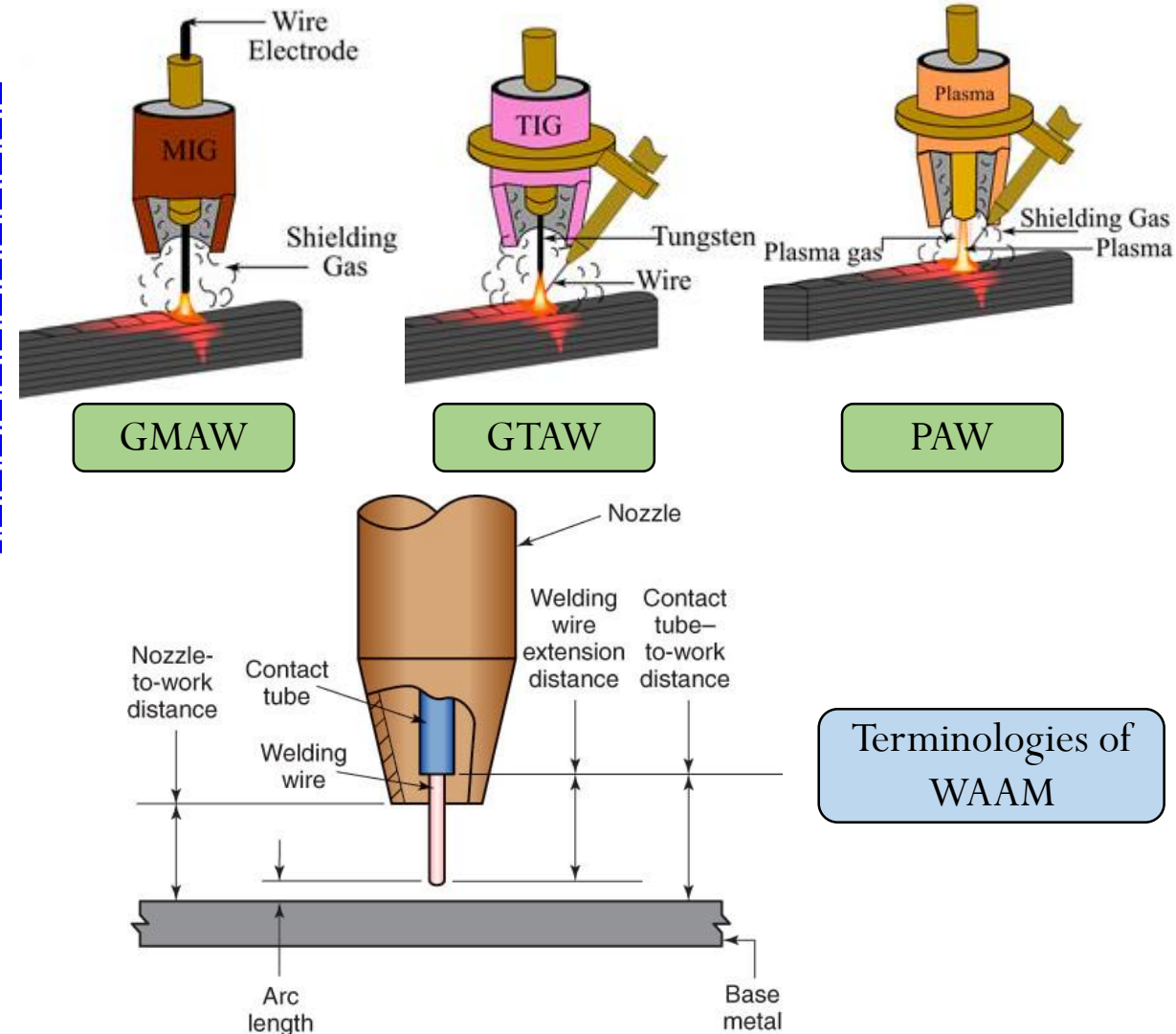


# Wire Arc Additive Manufacturing

## Setup

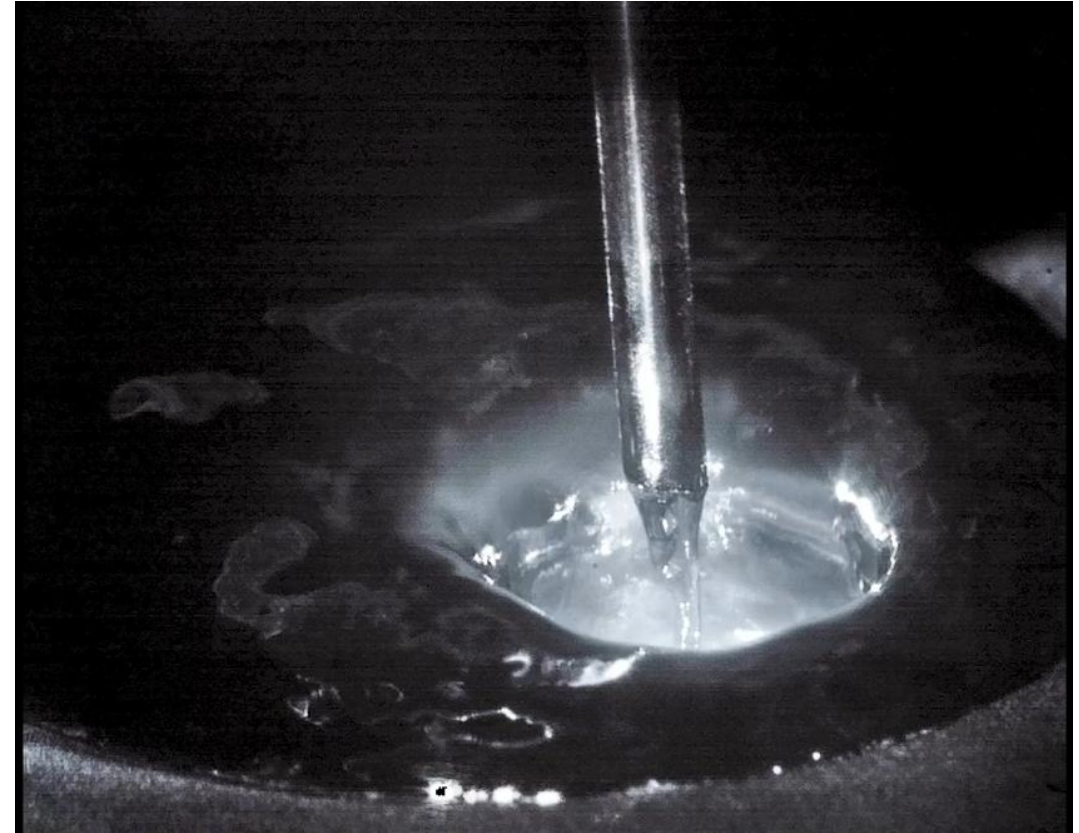
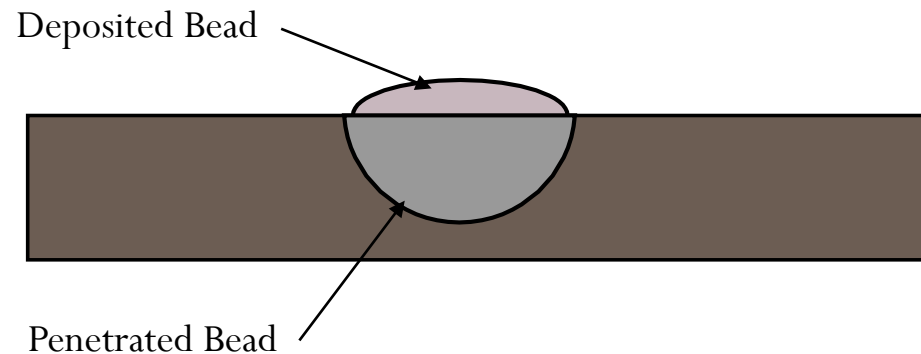
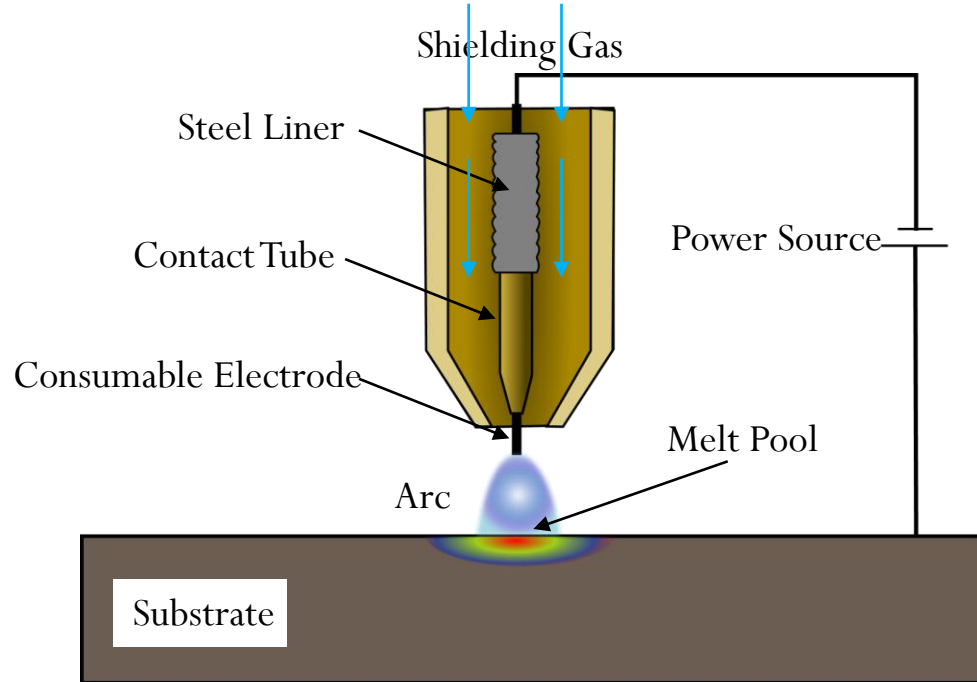


Low Carbon Steel, Stainless Steel, Tool steel(P91), Titanium alloys, Aluminum alloys, Inconel, stellate etc.



# Wire Arc Additive Manufacturing

## Gas Metal Arc Welding



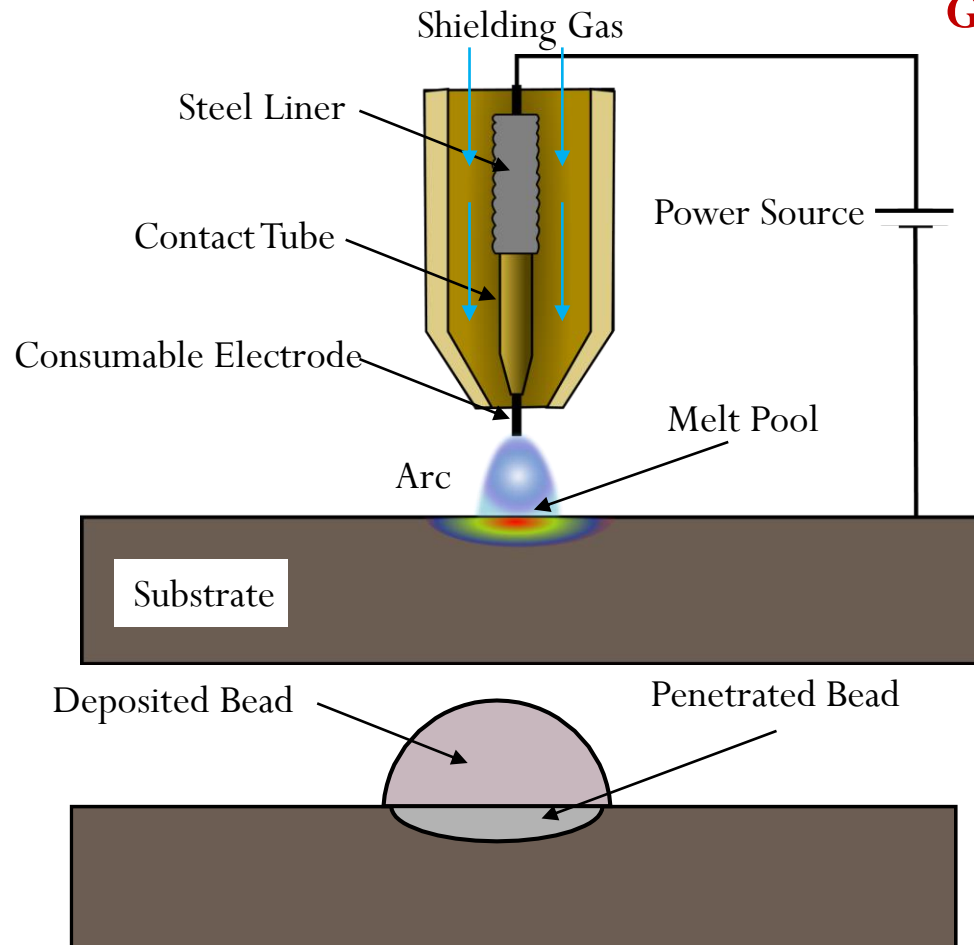
### Direct Current Electrode Negative

- Electrode is Positive
- $1/3^{rd}$  heat is generated on electrode
- $2/3^{rd}$  heat is generated on substrate



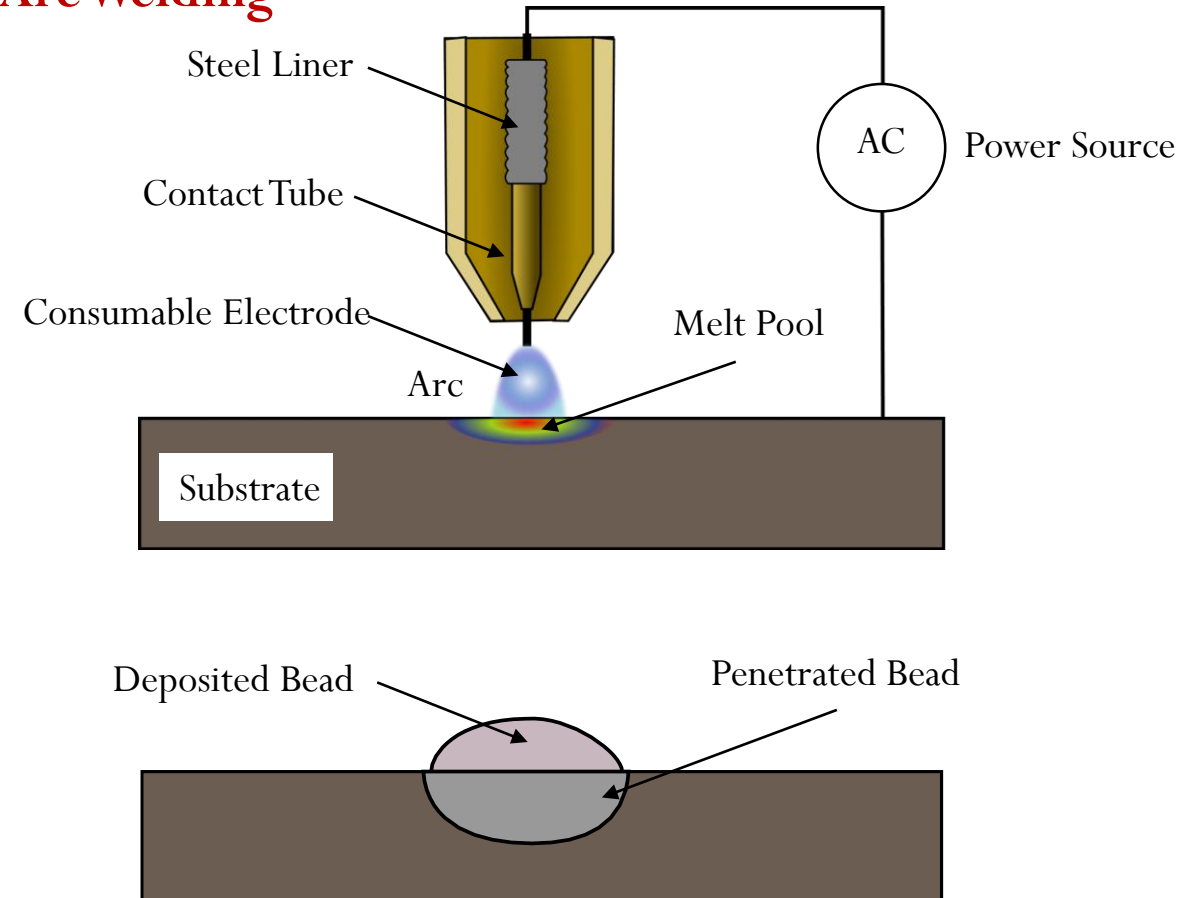
# Wire Arc Additive Manufacturing

## Gas Metal Arc Welding



### Direct Current Electrode Positive

- $1/3^{rd}$  heat is generated on Substrate
- $2/3^{rd}$  heat is generated on electrode

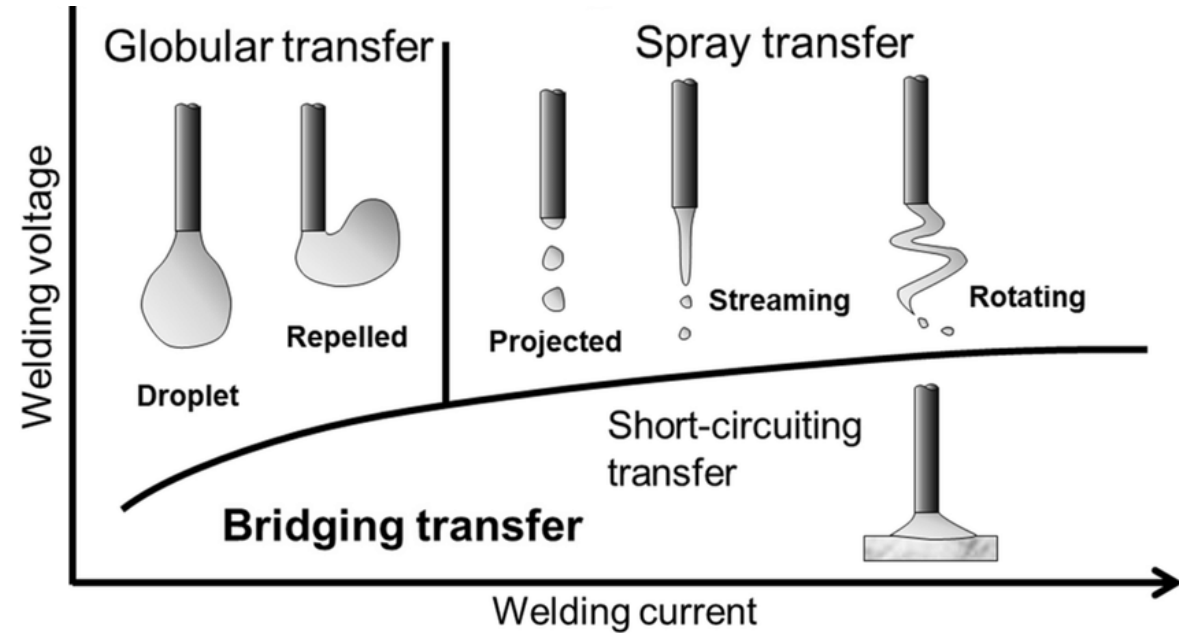
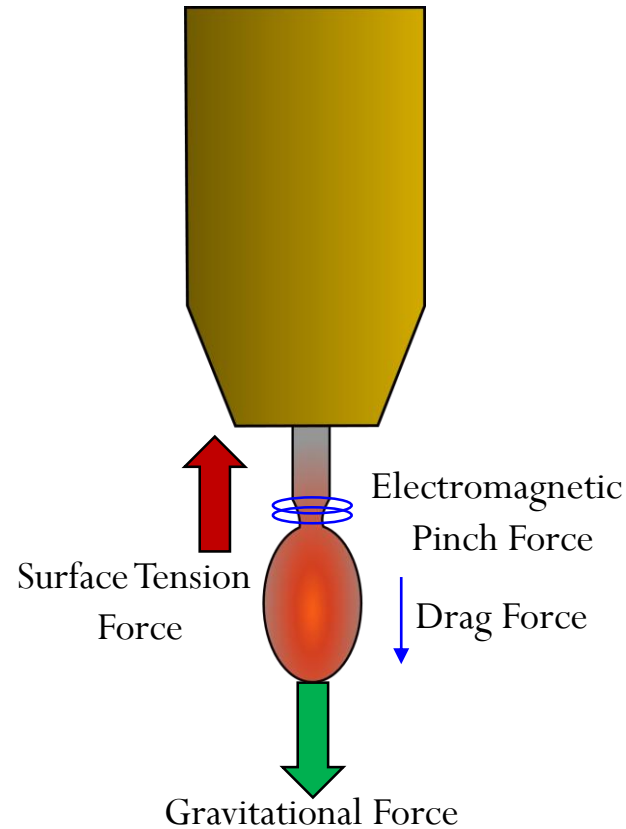


### Alternating Current Power source

- Equal heat is generated on both electrode and substrate at each half of the AC cycle

# Wire Arc Additive Manufacturing

## Gas Metal Arc Welding: Metal Transfer Modes



| Metal Transfer Mode | Deposition Rate (kg/hr) |
|---------------------|-------------------------|
| Short circuit       | 0.9 – 2.7               |
| Globular            | 1.8 – 3.2               |
| Spray               | 2.7 – 5.4               |
| Pulse Spray         | 0.9 – 2.7               |
| Rotating Spray      | 6.9 – 13.5              |

# Wire Arc Additive Manufacturing

## Gas Metal Arc Welding: Cold Metal Transfer

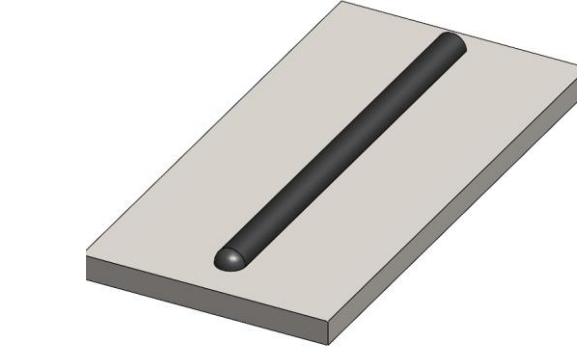
- Most popular method for WAAM due to its ability to precisely control the heat input and reduce spatter
- Invented by Fronius International GmbH, Austria in 2003
- Melt pool is hot when the arc is initiated and is cold as the arc is extinguished and the wire is retracted.
- Arc is initiated with main welding current and the feedstock wire is fed towards the weld pool
- As the droplet is formed, the feedstock wire is dipped into the melt pool. The welding current is reduced to small value
- Once the molten metal is transferred into the melt pool, the feedstock wire is retracted back and the main current is again supplied and the process is repeated.



# Wire Arc Additive Manufacturing

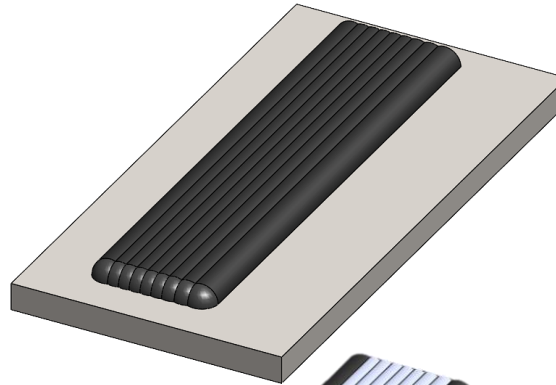
## Gas Metal Arc Welding: Process Parameters

| Parameters      | Variables                                                             |
|-----------------|-----------------------------------------------------------------------|
| Power           | Arc voltage, Starting current, Main Current and End current           |
| Filler          | Wire type<br>Wire material<br>Wire feed rate                          |
| Gas             | Gas Composition<br>Gas flow rate, gas pressure, Gas pre and post flow |
| Deposition      | Bead width<br>Bead height<br>Layer thickness                          |
| Deposition head | Stick out length<br>Standoff distance<br>Torch Speed                  |
| Toolpath        | Different area filling strategies<br>Effect stepovers                 |



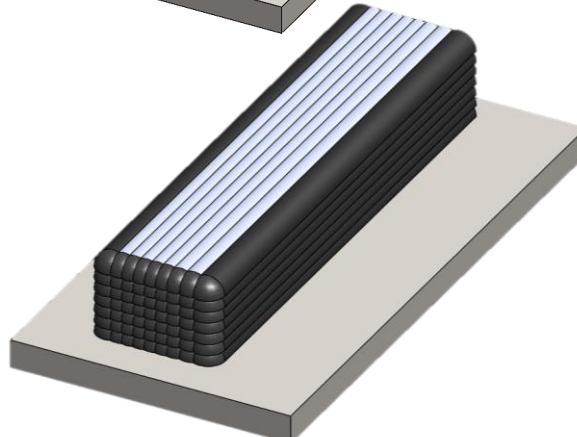
### Single Bead Experiments

- Bead penetration
- Bead height
- Bead Contact angle
- Bead Profile for different process parameter



### Multi Bead Experiment

- Step over between two beads
- Surface Flatness
- Different toolpaths
- Layer height

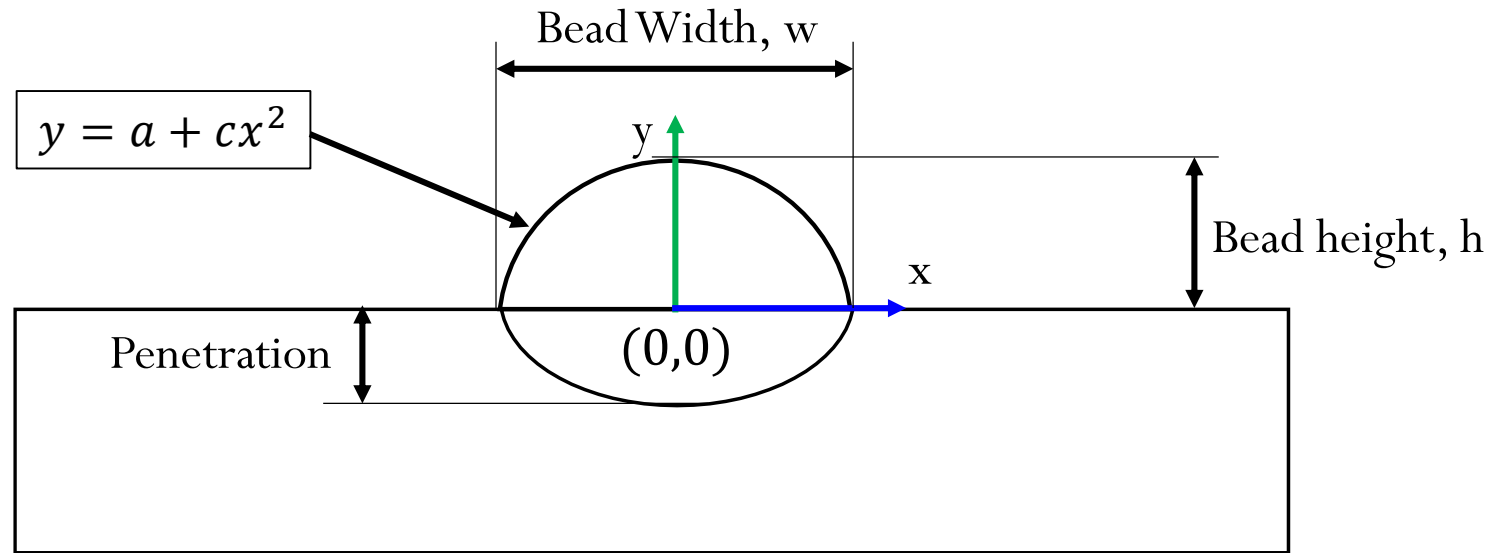


### Multi Layer Experiments

- Tensile Properties
- Wear Properties
- Hardness variation
- Fatigue and impact test

# Wire Arc Additive Manufacturing

## Gas Metal Arc Welding: Single Bead Experiment



- The modeling of the bead profile is required to establish the relationship among the different process parameters
- The bead profile above the substrate can be assumed to be a symmetric parabola of the form

$$y = a + cx^2$$

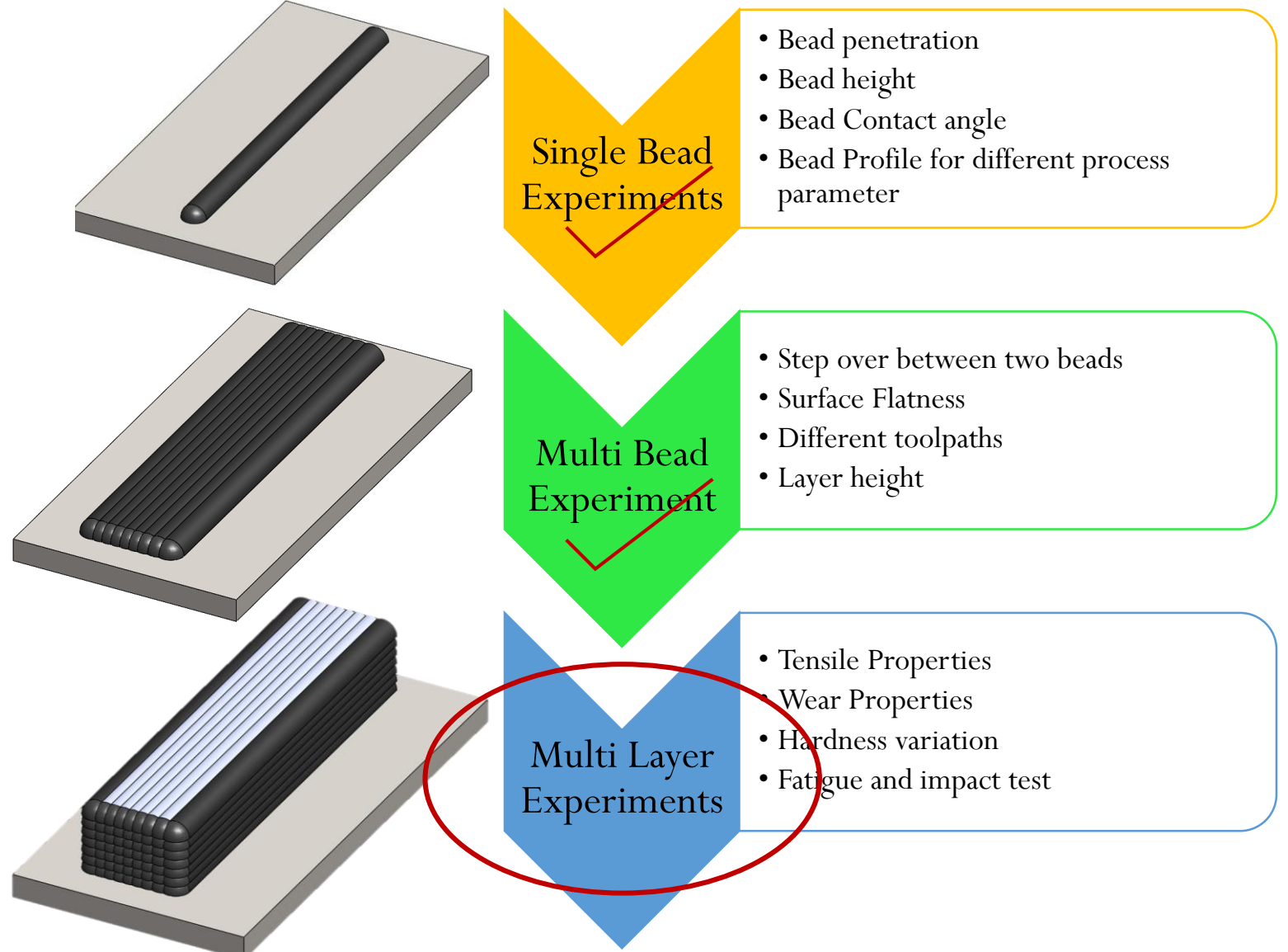
$$\text{Where, } a = h \text{ and } c = -\frac{4h}{w^2}$$



# Wire Arc Additive Manufacturing

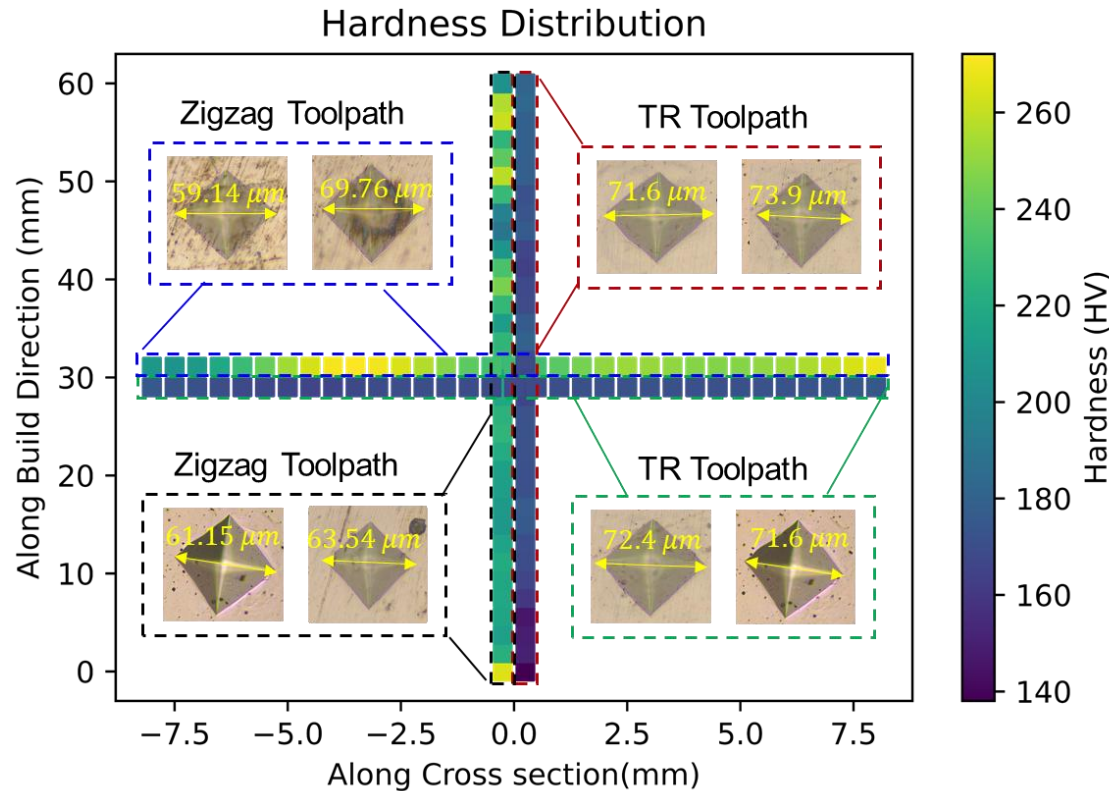
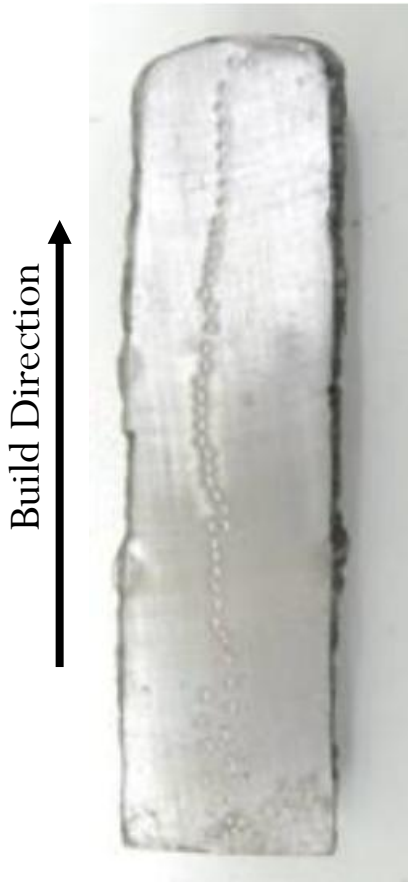
## Gas Metal Arc Welding: Process Parameters

| Parameters      | Variables                                                             |
|-----------------|-----------------------------------------------------------------------|
| Power           | Arc voltage, Starting current, Main Current and End current           |
| Filler          | Wire type<br>Wire material<br>Wire feed rate                          |
| Gas             | Gas Composition<br>Gas flow rate, gas pressure, Gas pre and post flow |
| Deposition      | Bead width<br>Bead height<br>Layer thickness                          |
| Deposition head | Stick out length<br>Standoff distance<br>Torch Speed                  |
| Toolpath        | Different area filling strategies<br>Effect stepovers                 |



# Wire Arc Additive Manufacturing

Gas Metal Arc Welding: Multi Layer: Low Carbon Steel

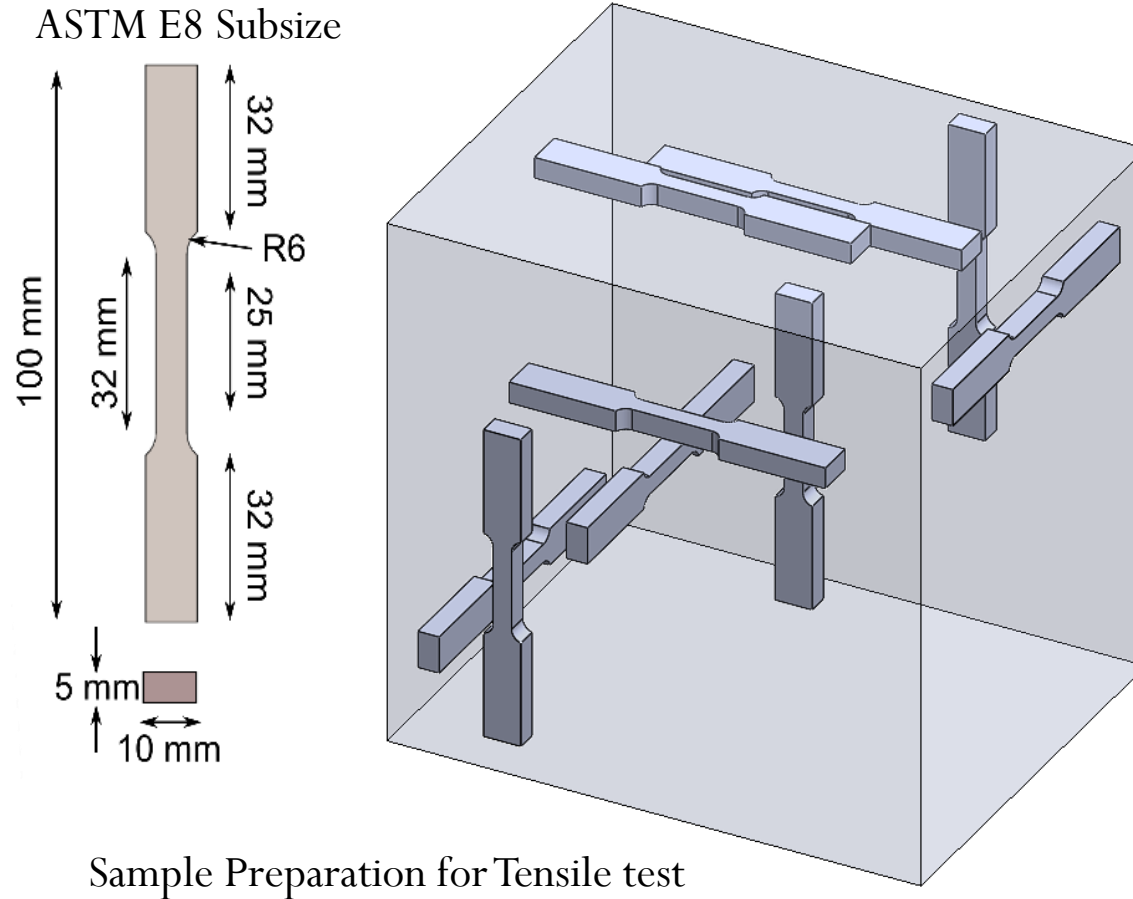


Hardness variation of a LCS deposited using WAAM

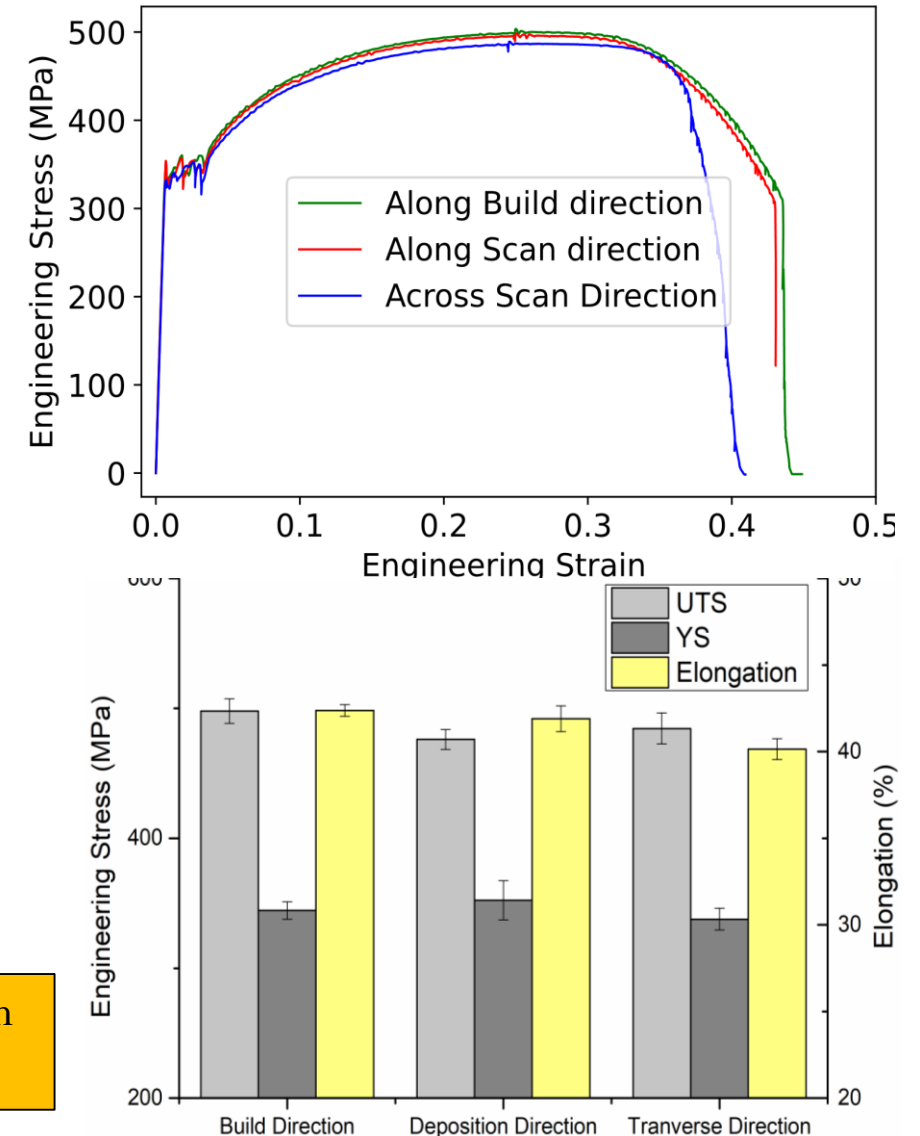
- During WAAM, the hardness variation depends upon the material deposited and the toolpath.
- Top layers often exhibit higher value of hardness as compared to other layers.
- The repetitive heating nature of the process results in lower hardness in the core of the deposited component.
- Toolpath followed during deposition plays key role in hardness variation across the deposition.
- For achieving uniform hardness, heat treatment is required to be performed on the fabricated part.

# Wire Arc Additive Manufacturing

Gas Metal Arc Welding: Multi Layer: Low Carbon Steel

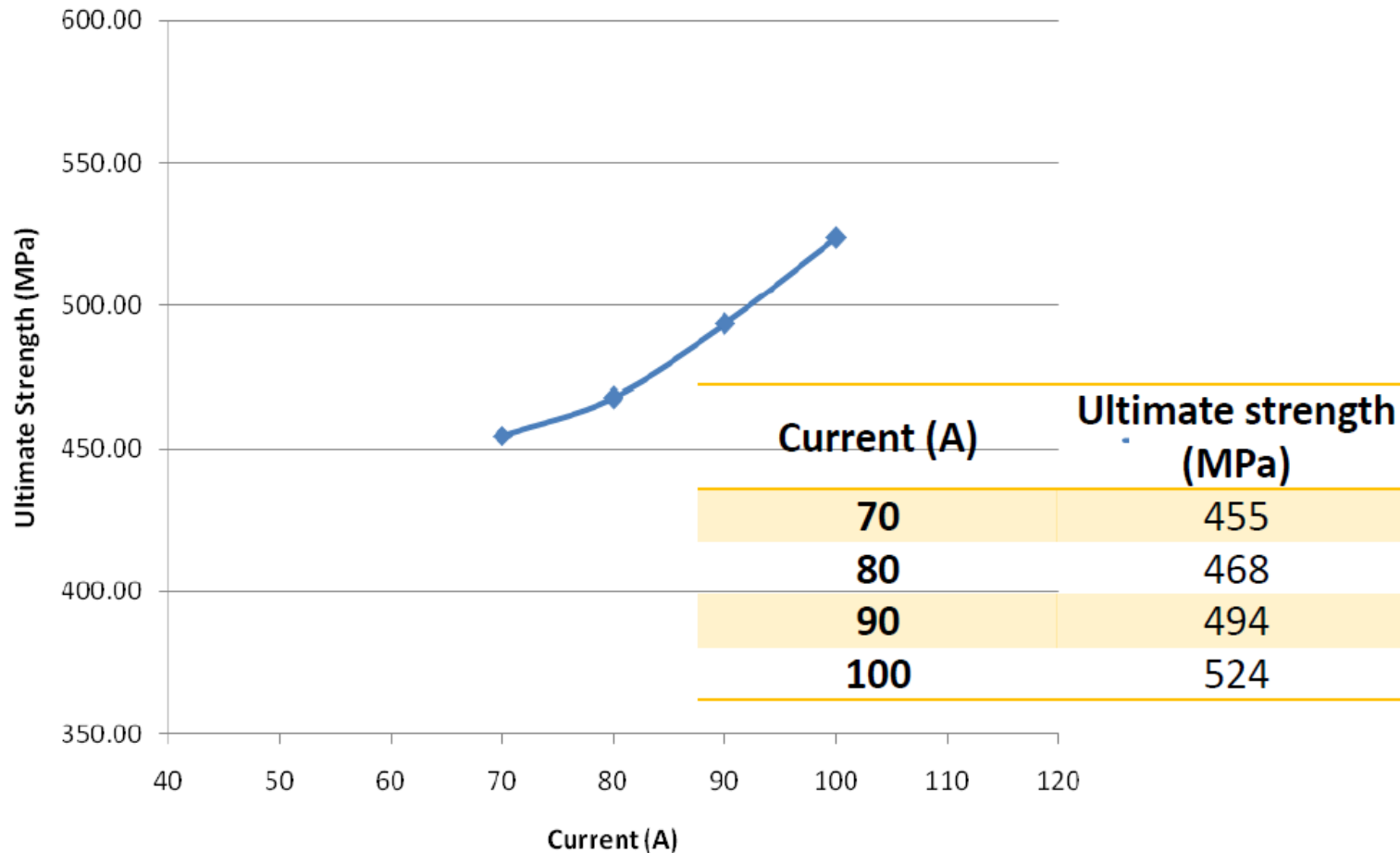


Samples along Deposition Direction, Build Direction and Transverse Deposition Direction should be taken out. Minimum three samples should be tested.



# Wire Arc Additive Manufacturing

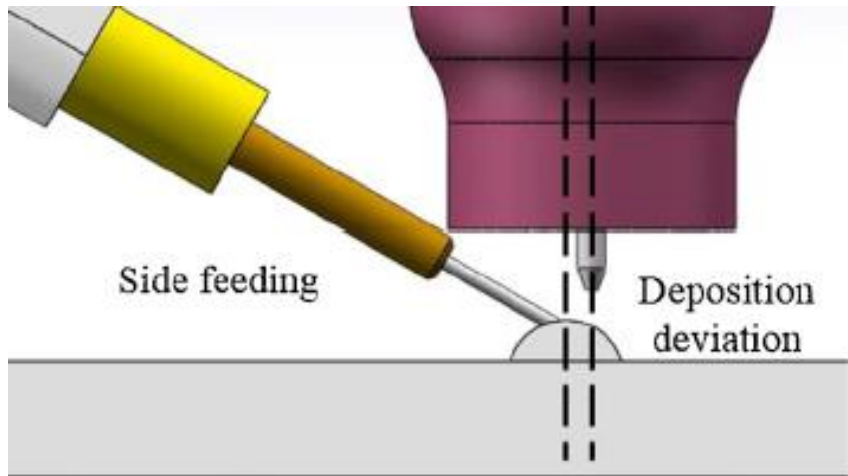
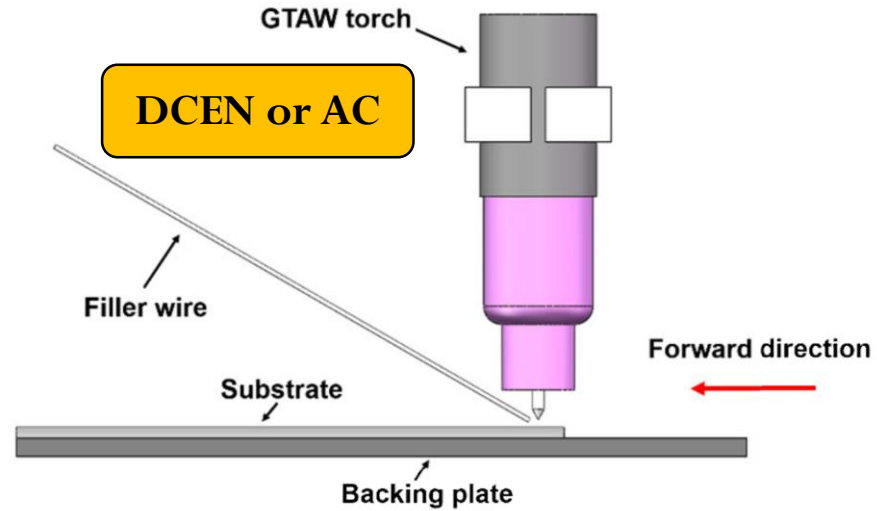
Gas Metal Arc Welding: Multi Layer: Low Carbon Steel



Tensile strength increase with increase in welding current due to increase in penetration of the deposited layer. However, a compromisation is required to be made with layer height and surface accuracy.

# Wire Arc Additive Manufacturing

## Gas Tungsten Arc Welding



Stainless Steel Deposition

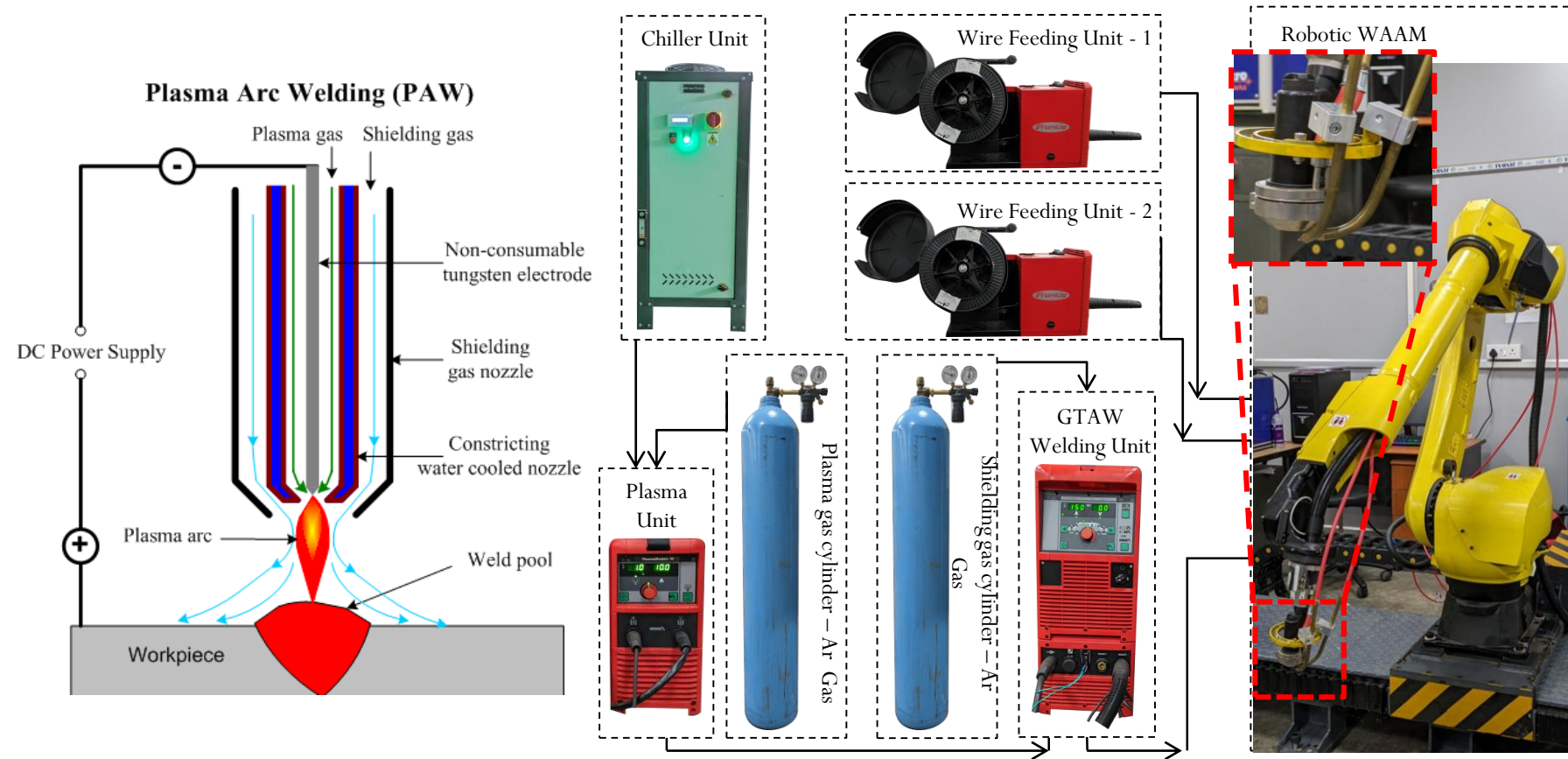
Titanium Deposition





# Wire Arc Additive Manufacturing

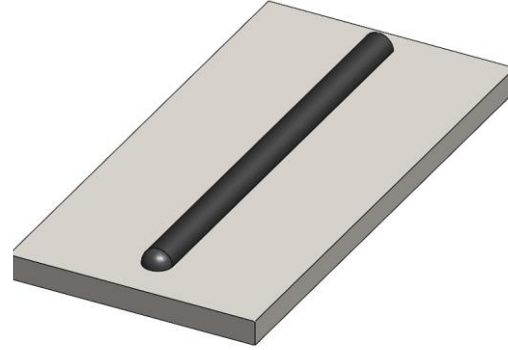
## Plasma Arc Welding



# Wire Arc Additive Manufacturing

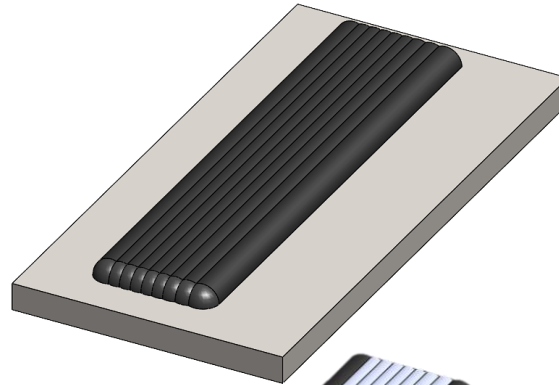
## GTAW: Process Parameters

| Parameters      | Variables                                                             |
|-----------------|-----------------------------------------------------------------------|
| Power           | Arc voltage, Starting current, Main Current and End current           |
| Filler          | Wire type<br>Wire material<br><b>Wire feed rate</b>                   |
| Gas             | Gas Composition<br>Gas flow rate, gas pressure, Gas pre and post flow |
| Deposition      | Bead width<br>Bead height<br>Layer thickness                          |
| Deposition head | Stick out length<br>Standoff distance<br>Torch Speed                  |
| Toolpath        | Different area filling strategies<br>Effect stepovers                 |



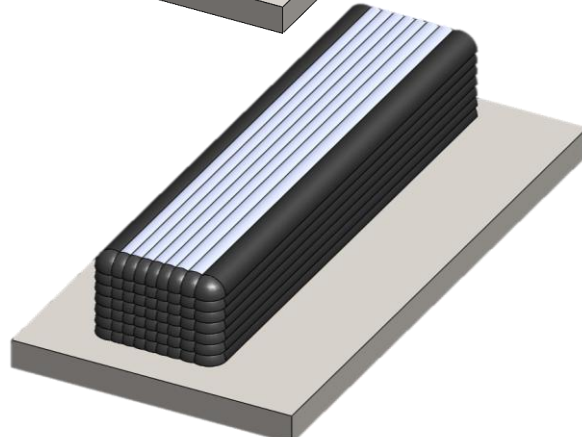
### Single Bead Experiments

- Bead penetration
- Bead height
- Bead Contact angle
- Bead Profile for different process parameter



### Multi Bead Experiment

- Step over between two beads
- Surface Flatness
- Different toolpaths
- Layer height

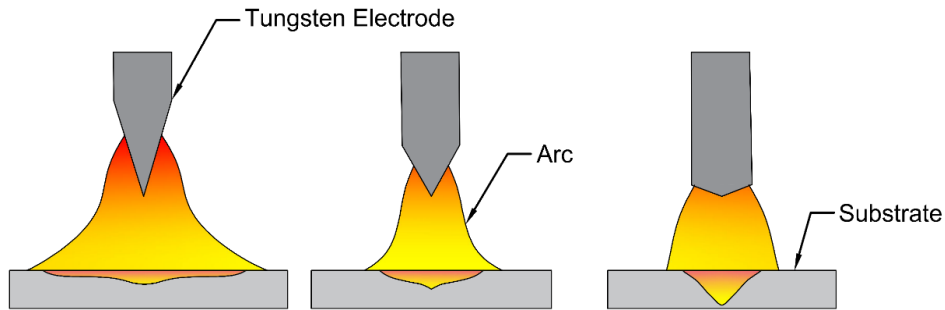
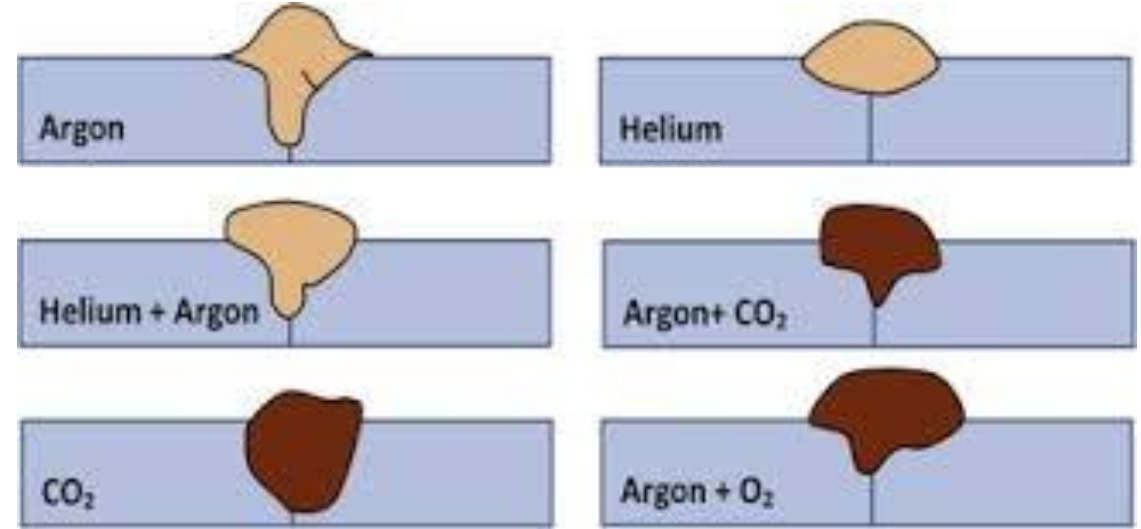
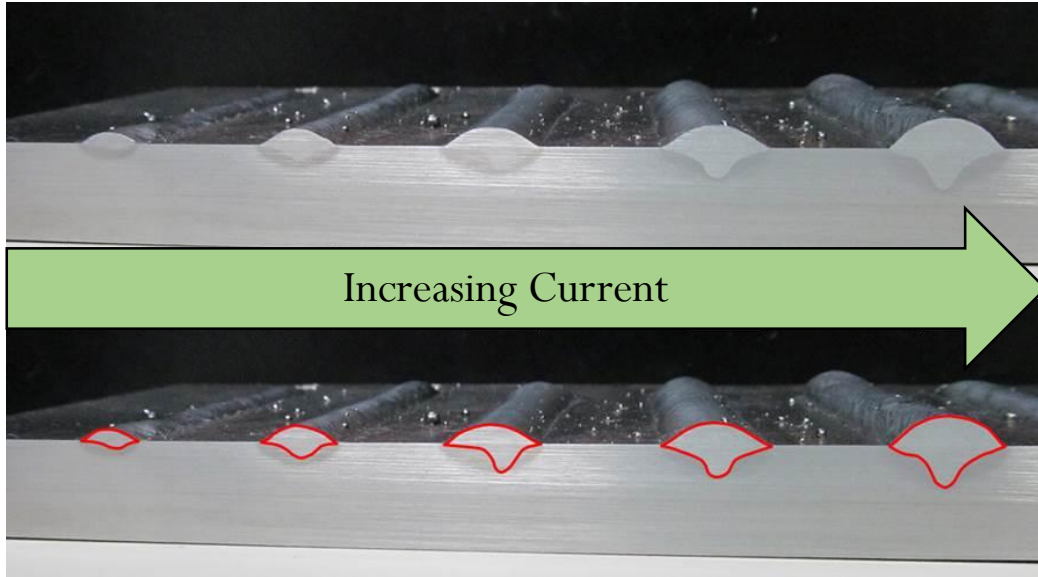


### Multi Layer Experiments

- Tensile Properties
- Wear Properties
- Hardness variation
- Fatigue and impact test

# Wire Arc Additive Manufacturing

## Effect of Different Parameters



30° Electrode Angle  
Wider Arc  
Less Penetration

60° Electrode Angle  
Less wider Arc  
Higher Penetration

140° Electrode Angle  
Straight Arc  
Highest Penetration

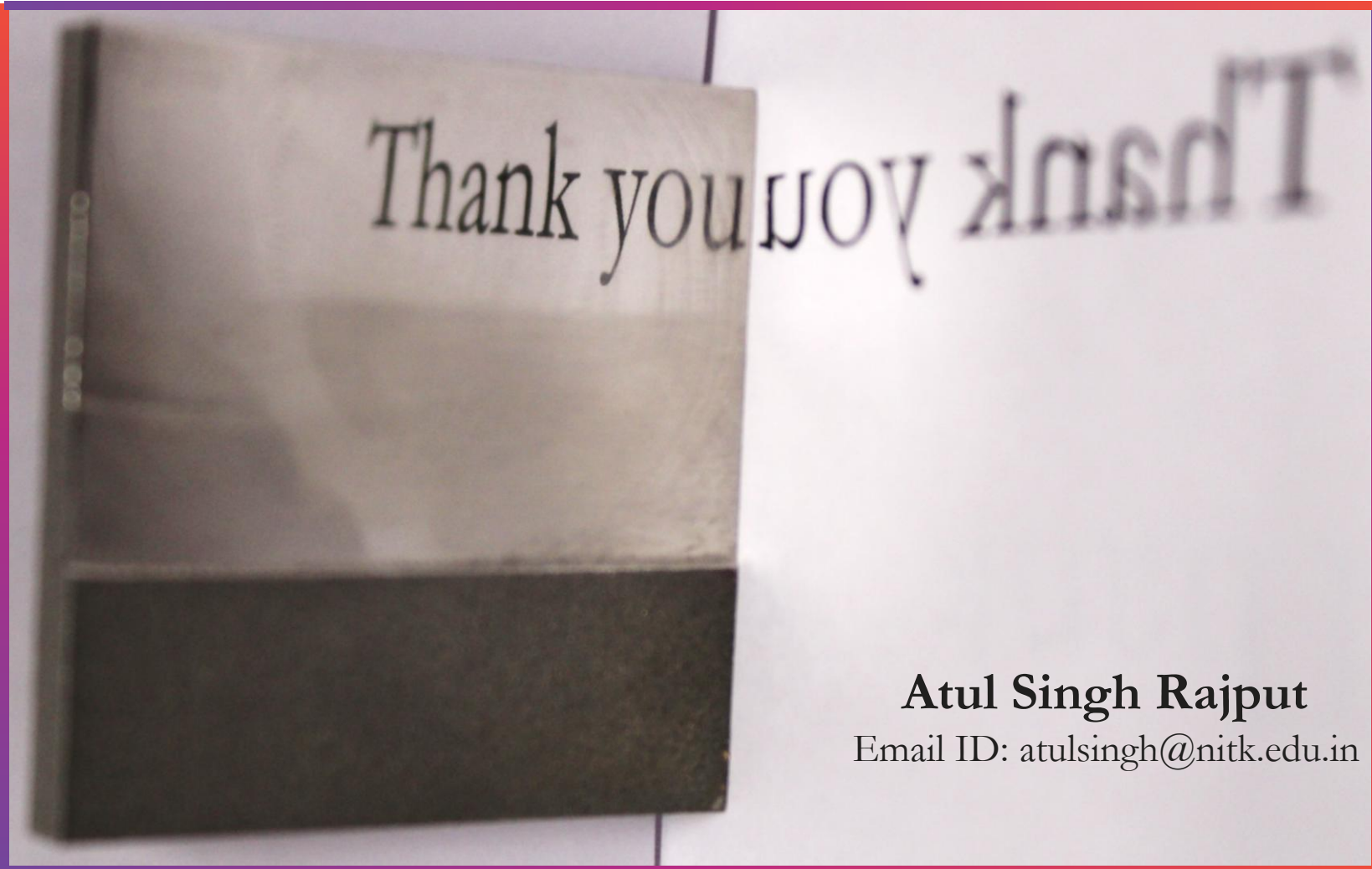
| Gas                           | Features                                                                                                                        |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Argon                         | Arc Stability; Smooth Stable weld bead; ideal for non ferrous materials for ability to reduce oxidation                         |
| CO <sub>2</sub>               | Less expensive, better penetration; More spatter                                                                                |
| Argon CO <sub>2</sub> Mixture | Good arc stability, penetration, and reduced spatter compared to pure CO <sub>2</sub>                                           |
| Helium                        | Higher thermal conductivity resulting higher heat input; with argon results higher penetration and heat input and welding speed |

# Wire Arc Additive Manufacturing

---

## Challenges in WAAM

- Achieving high **surface quality** in WAAM parts can be challenging due to factors such as uneven heat distribution, arc instability, and the buildup of spatter.
- Rapid heating and cooling during WAAM can lead to significant **residual stresses** and **distortion** in the manufactured parts, especially in large and complex geometries. Managing these residual stresses is crucial to ensure dimensional accuracy and mechanical integrity
- **Not all materials** are suitable for WAAM. The process requires wire feedstock with consistent composition and diameter, which limits the range of available materials. Furthermore, some materials may exhibit **poor weldability** or may be prone to **cracking during deposition**.
- Process optimization is necessary to minimize defects like **porosity, lack of fusion**, and **bead shape irregularities**.
- Complex geometries or overhanging features may require support structures to prevent sagging or distortion during deposition. Designing and implementing these **support structures without interfering** with the deposition process can be challenging and may increase material usage and post-processing requirements.
- As WAAM is a relatively new manufacturing technology, there is a **lack of standardized procedures** and **certifications** for WAAM-produced parts.



Thank you

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